



Fast Approaches for the Simulation of Advanced Rolling Processes

by

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Abstract

This thesis aims to investigate the potential of empirical and analytical models as an efficient alternative to finite element analysis simulations of advanced rolling processes. Research into simulations of advanced rolling processes shows that attempts have been made to improve the efficiency of finite element simulations but analytical and empirical models have been developed for both flat rolling processes and more advanced shape rolling processes. The rolling simulation tool, PyRoLL is introduced as a potential alternative. Four models are developed in PyRoLL consisting of a flat rolling model, a round-oval geometry model, a square-diamond geometry model, and finally a custom geometry based on a constricted box shape. These models are replicated in Abaqus except for the flat rolling model to be compared to PyRoLL and assess the accuracy of the results. In both simulation tools, the vertical roll force, outgoing profile width, and outgoing profile geometry are recorded for all models to be compared. Inconclusive results are obtained from the flat rolling simulations. The round-oval and square-diamond models produce similar results. PyRoLL predicts slightly larger roll forces than Abaqus (approximately 5-15% larger) and slightly larger outgoing widths (approximately 2% larger) for both of these models. PyRoLL predicts simplified output geometries with flat side edges which differ from experimental observations and FEA predictions. The constricted box geometry is more complex and PyRoLL predicts a slightly smaller outgoing width and a roll force that is less than half the force predicted by the FEA simulation. Looking deeper into the mechanisms of these discrepancies show that PyRoLL assumptions and simplifications are not suitable for more complex geometries as they are not representative of real world behaviours. Ultimately, PyRoLL has potential to be used for regular geometries such as the round-oval and square-diamond models but in its current state is not suitable for more complex shapes. PyRoLL can be used as an efficient alternative to FEA simulations but in some cases would require additional validation to confirm results. Further testing based on experimental data could provide additional validation and the confidence to use PyRoLL as the sole simulation tool as an efficient alternative to FEA simulations.

Statement of Original Authorship

I hereby certify that the submitted work is my own work, was completed while registered as a candidate for the degree stated on the Title Page, and I have not obtained a degree elsewhere on the basis of the research presented in this submitted work.

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Nomenclature

A – Initial yield stress

A_d – Contact area

A_{0L} – Incoming Lendl's area

A_{1L} – Outgoing Lendl's area

b – Width of cross section

b_0 – Equivalent incoming cross section width

b_1 – Equivalent outgoing cross section width

c – Wusatowski velocity coefficient

C_i – Zouhar contact coefficient

d – Groove depth

F_{Roll} – Roll force

G – Roll gap

h_0 – Equivalent incoming profile height

h_1 – Equivalent outgoing profile height

h_m – Mean height of workpiece

H_p – Height of pass

i – Indent

k_f – Yield strength

$k_{f, m}$ – Mean yield strength

k_m – Average deformation resistance

L – Projected contact length

L_d – Contact length

m_i – Freiberg flow stress exponents

P – Average contact stress

p – Roll face padding

Q_F – Inverse forming efficiency

R – Roll radius

r_i – Radius used in cross section definition

v – Workpiece velocity

w – Wusatowski spreading exponent
 w_g – Ground width
 w'_g – Even ground width
 w_L – Lendl's width
 w_t – Tip width
 w_u – Usable width
 α – Groove flank angle
 α_i – Angle used in cross section definition
 β – Ratio of outgoing workpiece width to incoming workpiece width
 β_n – Neutral line angle
 γ – Reduction in height of workpiece
 ϵ – Relative draught/reduction in height
 σ_R – Reverse tension
 σ_V – Forward tension
 φ – Strain
 $\dot{\varphi}$ – Strain rate
 ϑ – Temperature
 μ – Coefficient of friction
 Δh – Reduction in profile height

1. Introduction

1.1. Context

What is rolling

Rolling is the process of plastically deforming metal by passing it between rolls. The workpiece is subjected to high compressive stresses from the squeezing action of the rolls and to surface shear stresses as a result of the friction between the rolls and the metal. These frictional forces are also responsible for drawing the metal into the rolls (Dieter and Bacon, 1988). Rolling comes in many forms used for different applications but regardless of the rolling process considered, many parameters contribute to the final result.

Hot and cold rolling

The first distinction that can be made within rolling is between hot-rolling and cold-rolling. As the names suggest, these are rolling processes carried out at either high or low temperatures. There is no specific temperature that marks the changeover point between hot and cold rolling but instead it is dependent on the material being worked. Specifically, hot-rolling takes place above the recrystallisation temperature of the metal being worked while cold-rolling takes place below this temperature.

Hot-rolling can be beneficial as elevated temperatures soften the metal, making deformation easier and reducing stress on machinery and equipment, as well as allowing larger deformations to take place in single rolling passes. Hot-rolling can also influence the microstructure of the metal itself, resulting in different properties of the final product. It is very common for hot-rolling to be used during the initial rolling of large ingots into blooms or billets and sometimes for further processing into finished and/or semi-finished products such as metal plate, sheet, rod, bar, pipe, rails or structural shapes.

Cold-rolling has become very important in industrial settings due to its advantages in terms of dimensional tolerances and material properties. Cold-rolling can produce sheet, strip, and other metal products with much tighter dimensional tolerances and higher quality surface finish, making it a preferable option for finishing processes. Cold-rolling also has the added benefit of strengthening the material by strain hardening which is desirable for most rolled products on account of the temperature being too low for recrystallisation to occur.

Roll Mill Design – Layouts

Rolling mills can also come in many different shapes and sizes depending on the requirements and specific products to be rolled. The most basic layout is the two-high rolling mill, consisting of two identical rolls rotating in one direction through which the workpiece is passed. Productivity can be improved using a two-high reversing mill which allows workpieces to be passed back and forth through the rolls by reversing their direction of rotation. A further increase in productivity can be obtained using a three-high rolling mill. This consists of an upper and lower driven roll as before but with an additional unpowered roll in the middle which rotates by friction, allowing two workpieces to be passed through the roll stand at the same time in opposing directions.

Roll Mill Design – Roller Size

The actual selection of the rolls themselves is important for the outcome of a rolling process. Smaller diameter rolls can be used to decrease power requirements and achieve tighter tolerances but they can limit the amount of reduction obtained in a single pass. Typically, smaller rolls are used for finishing passes to meet tighter tolerances and these passes produce smaller reductions anyway, so this becomes less of an issue. Small rolls do have a tendency to deflect more than larger rolls so oftentimes when smaller rolls are used, they are backed up by larger rolls in layouts such as the four-high rolling mill, similar to the two-high rolling mill with a larger backup roll to provide extra support for each of the working rolls, and the cluster mill which utilises multiple levels of backup rolls to support the working rolls.(Wusatowski, 1969)

Roll Trains

A final product is nearly always produced over multiple roll passes, each gradually reducing the thickness of the material further, ending once the desired tolerances and surface finishes are met. For high production, it is common to implement sets of rolls, also known as stands, in sequence so that the workpiece can be passed directly from each stand to the next without manual input.

Shape Rolling

Another distinction that can be made is between flat rolling and shape rolling. Flat rolling is as described previously where the rolls are cylindrical in shape. Shape rolling is different in that the rolls are imprinted with grooves to produce the desired shape. The addition of grooves opens up a whole new world of possibilities for rolling metal products. Not only can metal be

rolled into basic geometries like sheets, strips, and basic bars, but they can be rolled into more complex shapes such as I-beams, channels, and railroad rails to name a few.

Some common groove shapes include round, oval, square, box, and diamond but many complex groove shapes are used for various applications and even within these basic groove shapes, there are many variations.

1.2. Motivation

Currently the most reliable way to simulate a shape rolling process of metal wires or bars is to use finite element analysis, otherwise known as FEA. This method of simulation consists of breaking up a continuous model into a finite number of parts or elements whose behaviour is specified by a finite number of parameters and the solution of the complete system as an assembly of its elements follows the same rules as those applicable to standard discrete problems. The distinct elements and nodes joining these elements can be represented in matrix form in terms of discrete nodal parameters that describe the behaviour of the elements and the system. (Zienkiewicz et al., 2025)

Although this is a very powerful and versatile tool that can provide very detailed outputs, it does come with downsides, mainly that it is very resource intensive to use. An FEA simulation can take hours or even days to run on a powerful computer with more complex simulations sometimes demanding weeks of simulation time depending on the computer specifications. Even before simulation, the models can take a long time to be developed as they must be built up from scratch. Models with more components, for example, rolling sequences with multiple passes consisting of different shaped rollers, require multiple rollers to be modelled and positioned correctly in the simulation. The time and financial requirements to run these simulations can be quite high when you consider both the amount of time that a trained engineer must spend developing the models and additional computing time that is required to actually run the simulation.

The nature and complexity of FEA simulations also adds to the costs and time requirements. An engineer must be very familiar with the software and also well versed in the specific area of application to produce meaningful and reliable simulations and outputs, but even the most skilled engineer will need to run multiple simulations. Due to the complexity of these models and simulations, it is unlikely that a complex simulation will be without error or produce the exact desired results on the first try, resulting in multiple simulations being required, taking hours each to run.

Although FEA simulations provide very detailed and important data, an alternative method of running simulations could greatly reduce the time and cost requirements of running simulations for a company. One possible approach is to use empirical solutions rather than finite element methods to run these simulations which could greatly reduce the model complexity and computing requirements and, as a result, greatly reduce the time and costs incurred by a company.

1.3. Overall Aim

As mentioned, the motivation for this project is to improve the efficiency of the simulation of advanced rolling processes. Despite efficiency and speed being the primary motivation, it is even more important that any results are suitable and representative of real world processes. The overall aim of this project then is to investigate the potential for analytical and empirical models to provide an efficient alternative to FEA simulations for advanced rolling processes while still producing results of high quality.

1.4. Specific Objectives

- Improve the efficiency of simulations of advanced rolling processes
- Maintain acceptable levels of accuracy in results
- Determine useful applications in industry or otherwise for fast acting simulations
- Identify limits for fast acting simulations

1.5. Thesis Layout

This thesis begins with an introduction to give context into what rolling is and introduce the problem to be investigated. This is followed by a literature review into the simulation of rolling processes and previous attempts to improve the efficiency of these simulations before introducing PyRoLL, a rolling simulation tool based on analytical and empirical models. The methodology consists of selecting models to create and simulate in PyRoLL, actually creating and running the simulations, and replicating these models in Abaqus, a finite element analysis software package that is used as a benchmark comparison to assess PyRoLL. Once the models are made and simulations run, the results are obtained and presented before being discussed. The discussion consists of comparing the PyRoLL results to experimental data, the FEA benchmark simulations, and finally proposing theories and evidence as to why the results differ and what that means for PyRoLL. Finally this paper ends with a conclusion to briefly tie together the various discussion points, give final thoughts on PyRoLL and make recommendations for how PyRoLL could be used and what direction could be taken if this project is continued.

2. Literature Review

2.1. Efficiency in Finite Element Simulation

Finite element analysis is a very powerful tool but despite its accuracy and versatility, it is very computationally demanding. Complex simulations offer a lot in terms of data and can save time and money by eliminating the need to build physical prototypes or carry out experimental testing in the early development stages of a project but, depending on the complexity, running a single simulation can range from hours to days and even weeks in certain cases. As a result, attempts have been made to develop FEA models in such a way as to reduce the computational demands. One study, although not related to metal forming, is based around the design and optimisation of electrical machinery and using FEA to do this. This study investigates two case studies, each consisting of a large number of iterations to optimise the design of electrical motors. The simulations for these case studies took 52 hours and 124 hours to simulate each. Although a large number of design iterations were tested for each case study (10,000 and 20,000 respectively), this is still 176 hours to simulate two different case studies (Sizov et al., 2013). Another study presents a method of FEA to simulate the stresses in a bridge. Despite being a more efficient method, this simulation still took 46 hours and 30 minutes to simulate overall, which is a long time for a single simulation (Markou and AlHamaydeh, 2024). These simulations taking days to complete shows the computational burden of FEA and in many industrial applications, simulations of these lengths just won't be feasible to complete.

Research has also been done to increase the efficiency of FEA simulations of metal rolling processes and multiple methods have been proposed and tested. One study proposed 6 different methods of simplifying a rolling model to reduce simulation time. The first of these methods is to utilise a non-uniform mesh density which can have a great impact on efficiency with minimal to no effect of the accuracy of the result if used correctly. The theory is that a dense mesh is used close to the deformation zone to maintain accuracy where deformation occurs but to use a more coarse mesh far away from the deformation zones, saving time by not simulating as many elements but not sacrificing the accuracy of results due to the fact that no deformation occurs here. The second method is to decrease the length of the rolled workpiece which once again simply requires less elements to be simulated. This study shortened the control workpiece of 500mm in length to 72mm. The risk with shortening the workpiece too much is that it may prevent the model's ability to achieve steady state rolling and affect the results. This is

considered for all the models developed during this study however and steady state is achieved in all simulations. (Gornyakov et al., 2021)

Two more modifications are suggested which reduce the number of elements to be simulated by making assumptions that simplify the theory behind the calculations. The first of these is to assume a rigid roller. In reality the roller is not rigid and will deform some amount but this theory assumes that the deformation is negligible and therefore can be ignored. This allows an analytical rigid roller to be modelled instead of a deformable roller which reduces the number of elements to simulate and also the computational power required to perform the calculations. The second method is to create a 2-dimensional rather than 3-dimensional model. This is only possible for simple models such as flat rolling but when applicable they can drastically decrease the number of elements to be simulated. This modification is based on the assumption of plane strain and plane stress theories and although proving very efficient, this study found the 2-dimensional model to be inaccurate and therefore not a suitable method of decreasing simulation time. (Gornyakov et al., 2021)

Finally, two methods are suggested in which different methods of solving the problem are investigated. As a comparison to the standard implicit solver analysis, tests were done with the explicit solver in Abaqus and also with a Eulerian steady state model. Altogether 5 simulations were completed consisting of a benchmark control model with none of the aforementioned efficiency measures and 4 efficient models, being the implicit 3D model, explicit 3D model, Eulerian mass flow model and a 2D implicit model. All these efficient models utilised the non-uniform mesh density, rigid roller, and shorter workpiece. (Gornyakov et al., 2021)

The time savings that came as a result of these measures were very noticeable. The control simulation with no time saving measures took 64,002 seconds (approximately 17.8 hours). The four efficient models all produced time savings of over 95% with the Eulerian steady state model taking the longest of these at 2,613 seconds (approximately 44 minutes). The efficient implicit and explicit models were very close in time taking 37 and 39 minutes to complete the simulation, time savings of 96.50% and 96.32% respectively. The 2D model yielded the most significant improvement with a simulation time of 302 seconds (just over 5 minutes), equivalent to a 99.5% reduction in simulation time. (Gornyakov et al., 2021)

2.2. Analytical & Empirical Models in Rolling

While improvements are being made in the efficiency of FEA simulations, they are still computationally and resource intensive. An alternative to these numerical models, empirical

models can be used to run simulations almost instantly on desktop computers or even laptops with relatively standard specifications. A number of models and formulae have been developed to predict particular parameters of rolling processes based on values that require little to no calculation to obtain.

One parameter of interest is the geometry of the outgoing workpiece, particularly the cross-sectional profile. Different shapes and products will have different requirements in terms of cross-section but one of the more common metrics is the width of the outgoing profile. One study develops a relationship to predict the width of contact area and the spread of wire in flat rolling processes when the incoming shape is round in cross-section using deformation pattern idea and examining FEM simulations. The FEM simulations show shear bands form in an X shape on the cross section. Initially, the bands are assumed to be inscribed vertically through the centre of the undeformed cross-section and as the height is reduced, they are rotated in opposite directions while the overall length of the bands remains unchanged. Fig.1 illustrates this mechanism with the bands labelled CC' and DD' being fitted on each other before deformation and rotating around the centre point as height is reduced further. (Kazeminezhad and Karimi Taheri, 2005)

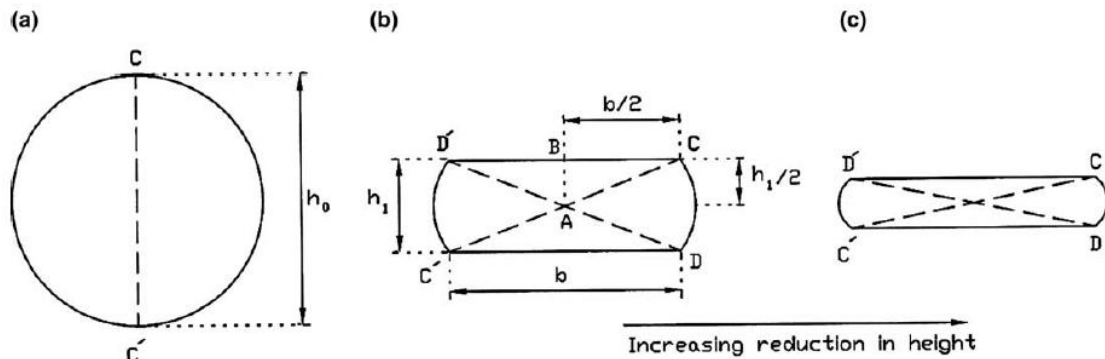


Figure 1: Mechanism of bands rotating during flat rolling

Both shear bands are equal in length to the diameter of the original cross-section, given by h_0 . Defining h_1 as the height of the reduced cross section and b as the contact width of the outgoing workpiece, a relationship can be drawn in the region ABC from Fig.1 (b) above using basic trigonometry. After simplification we arrive at equation (1).

$$b = \sqrt{h_0^2 - h_1^2} \quad (1)$$

We can also define the term Δh as the change in height of the cross section which is equal to $h_0 - h_1$. Using this definition and some basic algebra, Eq.1 can be rewritten as

$$b = \sqrt{\Delta h(2h_0 - \Delta h)} \quad (2)$$

Up until this point, it was assumed that the length of these bands remained consistent when in reality they increase slightly. Assuming that this increase in length is equal to $\Delta h/2$, Eq.2 can be simplified and rewritten as

$$b = \sqrt{2h_0\Delta h} \quad (3)$$

It is observed that the contact width b is proportional to the square root of the reduction in height of the wire. To verify this relationship, test passes were run under different conditions. Wires made of 5.5mm diameter low carbon steel, 5.5 and 13mm high carbon steel and 8mm copper wire were rolled with different reductions in height and roll speeds. The results showed that the contact width is not sensitive to roll speed and that material properties do not play a part either as the high and low carbon steels of same diameter produced the same results. The high and low carbon steel results can be seen in Fig.2 and Fig.3.

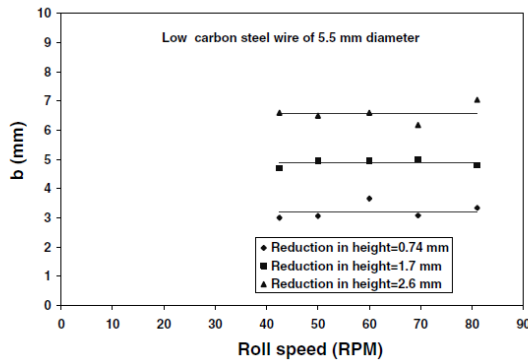


Figure 2: Spread of low carbon steel wire with respect to roll speed and reduction in height

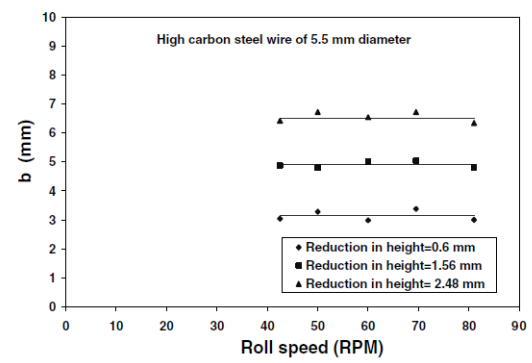


Figure 3: Spread of high carbon steel wire with respect to roll speed and reduction in height

As seen in the derivation of Eq.3 and the subsequent validation tests, empirical models can be quickly developed from previous simulation and/or experimental data and validated equally quickly. Despite rolling being a very complex process with many parameters, the contact width in flat rolling can be described with a very simple relationship with the incoming diameter and reduction in height.

2.3. PyRoll as an Empirical Analysis Tool

Developing analytical relationships to describe the various parameters of rolling processes is one thing but implementing them to design and analyse actual rolling processes in rolling mills

is a different story. Attempts have been made to apply these models to software programmes in the form of analytical models and hybrid models consisting of FEM and analytical components. One such tool called PyRoLL is discussed below. This tool combines a number of individual analytical and empirical models for specific rolling parameters into one tool in order to perform extensive simulations of advanced shape rolling processes.

Basic PyRoLL Model Creation

PyRoLL is an empirical analysis tool that may provide a time and cost-efficient alternative to FEA. PyRoLL comes in the form of an open source Python plugin package which can be used to create rolling models through interchangeable modules, either based on premade templates or from the ground up for more customisable designs. It doesn't produce 3-dimensional models but instead performs calculations based on a steady state process based on cross-sectional profiles of incoming material, rollers, material properties and a number of other aspects. This results in simulations which can be completed in a manner of seconds as analytical and empirical equations are solved very quickly rather than simulating thousands or even millions of elements for a large number of iterations as is the case in finite element methods. ("Home — PyRoLL 2.1.1 documentation," n.d.).

The modules required for a PyRoLL simulation are an incoming profile, a groove geometry, a roll design, and a roll pass design at a minimum. The profile module is used to define the incoming material stock with both geometrical and material properties. There are a number of basic profile shapes such as round, box-shaped, diamond, and hexagonal to name a few which only need a small number of inputs to fully define their geometry. For example, the box profile requires the height, width, and corner radius to be input while the round profile only requires either a radius or diameter. Once the geometry has been defined, the profile can be created after the material properties have been assigned. These material properties are the density, specific heat capacity, flow stress (which is equivalent to yield stress), and finally the temperature of the incoming material.

The groove module defines the pattern/geometry that will be inscribed onto the rollers. There are multiple groove classes with some having variations allowing more detail in the design. The box-like groove class, for example, can produce a simple box groove to produce products of rectangular cross section but also provides an option for a constricted box groove which allows the user to specify an indent in the groove, producing a wider variety of product geometries. There are multiple variations within the different groove classes similar to the two

varieties of box groove mentioned. All of the groove geometries available are based on a generic elongation groove consisting of two straights and four radii (Fig.4). The actual groove types available to use are special cases of this generic elongation groove with some measures being unequal and some being equal to zero. The measures given in this diagram are listed below in table 1.

Table 1: Symbols and descriptions of components in the generic elongation groove

Symbol	Description
r_i	Radius
α_i	Angle corresponding to a radius
α	Groove flank angle
β, γ	Angles useful for coordinate calculation
φ	Roll face angle
w_g	Ground width
w'_g	Even ground width
w_t	Tip width
w_u	Usable width
d	Depth
i	Indent
s	Roll gap
p	Roll face padding

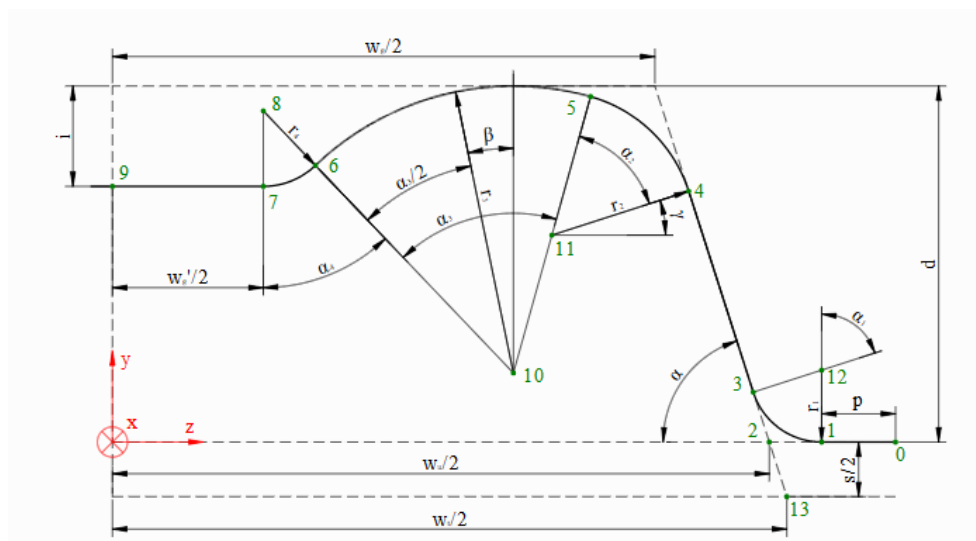


Figure 4: Generic elongation groove geometry used as base for PyRoLL cross sections

The co-ordinates of the points 0-12 can be calculated based on formulas given in table 2 below.

Table 2: Formulae to calculate the coordinates of points in the generic elongation groove

Number	z-coordinate	y-coordinate
0	$z_1 + p \cos \varphi$	$z_1 + p \sin \varphi$
1	$z_2 + r_1 \tan \frac{\alpha_1}{2} \cos \varphi$	$r_1 \tan \frac{\alpha_1}{2} \sin \varphi$
2	$\frac{w_u}{2}$	0
3	$z_{12} - r_1 \sin \alpha$	$y_{12} - r_1 \cos \alpha$
4	$z_{11} + r_2 \cos \gamma$	$y_{11} + r_2 \sin \gamma$
5	$z_{10} + r_3 \sin \left(\frac{\alpha_3}{2} - \beta \right)$	$y_{10} + r_3 \cos \frac{\alpha_3}{2}$
6	$z_8 + r_4 \sin \alpha_4$	$y_8 - r_4 \sin \alpha_4$
7	$\frac{w'_g}{2}$	$d - i$
8	$\frac{w'_g}{2}$	$y_7 + r_4$
9	0	y_7
10	$z_6 + r_3 \sin \left(\frac{\alpha_3}{2} + \beta \right)$	$y_6 + r_3 \cos \left(\frac{\alpha_3}{2} + \beta \right)$
11	$z_{10} + (r_3 - r_2) \sin \left(\frac{\alpha_3}{2} - \beta \right)$	$y_{10} + (r_3 - r_2) \cos \left(\frac{\alpha_3}{2} - \beta \right)$
12	$z_1 - r_1 \sin \varphi$	$y_1 + r_1 \cos \varphi$

Once the incoming profile and groove geometry have been defined, a roll is defined with a radius or diameter and a previously defined groove shape. This creates a roller which is used alongside a roll gap to define a roll pass through which the workpiece is passed. The simulation is initiated by applying the solve function to the roll pass with the incoming workpiece profile taken as an input.

For more complex simulations, multiple roll passes can be defined and applied to a sequence where the workpiece is directly fed from one set of rolls to the next with the possibility to specify transport sections in between rolls to simulate thermal properties such as cooling to the atmosphere.

PyRoLL Plugin Packages

The PyRoLL core package uses various methods to calculate the properties of the outgoing workpiece, which are useful for quick simulations but this core package is lacking in a number of areas. For a more detailed analysis, additional plugins are available which utilise more advanced models and provide different results, accounting for real world phenomena that occur during rolling processes. The code used to implement these models and documents describing the models are available on the PyRoLL GitHub with a repository for each relevant plugin (“pyroll-project repositories · GitHub,” n.d.).

Wusatowski Spreading Model

The core PyRoLL package is very poor at predicting the width of the outgoing profile as it doesn't use a model to calculate the width, instead assuming that it is equal to the usable width of the groove shape.

The Wusatowski Spreading Model plugin is utilised to calculate the width based on the height reduction of the workpiece and correction coefficients rather than assuming it is exactly the same as the groove through which it is passed. The Wusatowski model is given below in Eq.4

$$\beta = a \times c \times d \times f \times \gamma^{-w} \quad (4)$$

The term β in Eq.4 is a ratio representing the spread of the workpiece, equal to the outgoing width of the workpiece divided by the incoming width. The calculation of the outgoing width is implemented in PyRoLL with a modified version of this equation, given in Eq.5, where the terms b_0 and b_1 represent the equivalent incoming width and equivalent outgoing width respectively.

$$b_1 = (a \times c \times d \times f \times \gamma^{-w}) \times b_0 \quad (5)$$

The terms a , c , d , and f are correction coefficients for temperature, velocity, material, and friction respectively. The term γ represents the reduction in height of the workpiece and is calculated by dividing the equivalent outgoing workpiece height (denoted h_1) by the equivalent incoming workpiece height (denoted h_0). The term w is the spread exponent, calculated from geometries of the workpiece and work rolls.

The spread exponent w is calculated using one of two equations which are derived from plastic stress-strain equations given by Mises. Wusatowski proposed Eq.6 for this exponent but adjusted it to give Eq.7 for large strains, specifically when the strain is greater than 50%. In

these equations, h_0 and b_0 represent the equivalent incoming height and width as previously defined while R represents the roll radius.

$$w_{low} = 10^{-1.269\left(\frac{h_0}{2R}\right)^{0.556} \frac{b_0}{h_0}} \quad (6)$$

$$w_{high} = 10^{-3.457\left(\frac{h_0}{2R}\right)^{0.958} \frac{b_0}{h_0}} \quad (7)$$

The coefficients a , c , d , and f as mentioned in Eq.5 are correction coefficients. The velocity coefficient c is calculated based on Eq.8 where v represents the velocity of the workpiece. The other correction coefficients are not automatically calculated by PyRoLL and instead are set to values of 1 by default so that they have no effect on the predicted spread. If data is available, these coefficients can be manually calculated or directly input by the user.

$$c = (-0.002958 + 0.00341\gamma)v + 1.07168 - 0.10431\gamma \quad (8)$$

Lendl's Equivalent Rectangle Model

It is noted that the Wusatowski spreading model described above doesn't use the maximum or average heights and widths but rather equivalent dimensions. This refers to the use of an equivalent rectangle method of predicting the cross sections. PyRoLL doesn't directly calculate the spreading of the defined geometry but instead estimates an equivalent rectangle which is representative of the actual geometry but allows flat rolling spreading models to be utilised. This package implements an equivalent method proposed by Lendl to simplify the calculations.

The idea behind this method is to define a contact width of the incoming profile and rolls by the intersection of their contours. This is visualised in Fig.5 with an example of a round cross section passing through an oval shaped groove.

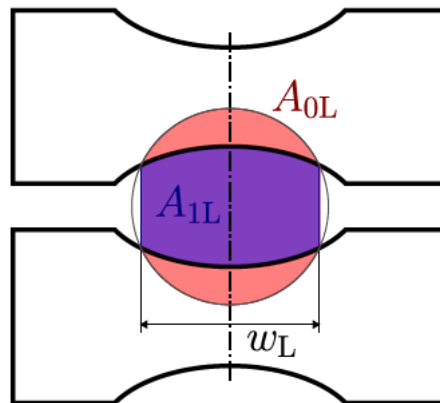


Figure 5: Diagram of Lendl's width and areas in the equivalent rectangle estimation method

The value w_L is the Lendl's width and is equal to the width of the intersection points between the workpiece and groove shape contours as shown in Fig.5. The values A_{0L} and A_{1L} are the Lendl's areas and are taken as the remaining cross sectional profiles after clipping the sections outside of the Lendl's width. The equivalent incoming area A_{0L} is equal to the entire incoming cross section less the clipped outer sections while the equivalent outgoing area A_{1L} is equal to only the groove profile inside this clipped width. This is visualised once again in Fig.5 where A_{0L} is equal to the sum of the blue and red areas while A_{1L} is equal to only the blue area confined by the groove geometry.

Once the Lendl's width and areas are found, the height of the equivalent flat pass is given by Eq.9 for both the incoming and outgoing profile.

$$h'_i = \frac{A_{iL}}{w_L} \quad (9)$$

This equivalent rectangle method can be used to calculate the width of the outgoing cross section but in the case of the PyRoLL Basic package, the equivalent heights and equivalent incoming width are taken as inputs for the Wusatowski spreading model to predict the width of the outgoing profile.

Freiberg Flow Stress Model

This package allows empirical flow stress models to be defined and implemented with several coefficients based on common shapes of flow stress curves in hot and cold forming. The flow stress in this model is assumed to be only influenced by temperature, strain, and strain rate. The formula for the Freiberg Flow Stress calculation is given in Eq.10.

$$k_f(\varphi, \dot{\varphi}, \vartheta) = A e^{m_1 \vartheta} \vartheta^{m_9} \varphi^{m_2} e^{\left(\frac{m_4}{\varphi}\right)} (1 + \varphi)^{m_5 \vartheta + m_6} e^{m_7 \varphi} \dot{\varphi}^{m_3 + m_8 \vartheta} \quad (10)$$

In this formula, k_f is the yield stress after being subjected to given strain, strain rate, and temperature represented by φ , $\dot{\varphi}$, and ϑ respectively. Once these inputs are known, the yield stress is calculated based on the material coefficients. These coefficients are, A , which represents the undeformed yield stress of the material and coefficients m_1 - m_9 , which are exponential coefficients relating to the aforementioned strain, strain rate, and temperature. A limited number of materials are available in PyRoLL which can be selected with all of these coefficients pre-defined but if sufficient data is known, a material can be created in PyRoLL by defining these coefficients to match the real world flow stress curves of the relevant material. For custom materials, any number of the exponent coefficients can also be ignored if they have

no effect by setting them to zero which eliminates their effect on the yield stress. This also means by setting all of these exponents to zero, a constant yield stress can be specified.

Lippmann-Mahrenholz Force and Torque Model

This plugin provides the roll force and torque model developed by Lippmann and Mahrenholz. The equations are derived from classic strip theory with simplifications for hot rolling and can be applied to shape rolling when an equivalent rectangle approach is used like in PyRoll. The roll force is given by Eq.11 below.

$$F_{Roll} = A_d k_{f,m} Q_F \quad (11)$$

In this equation, the term A_d represents the contact area between the roll and workpiece, $k_{f,m}$ represents the mean flow stress of the material, and Q_F represents the inverse forming efficiency. The mean stress is calculated as a weighted average of the incoming and outgoing yield stresses with a 2:1 ratio of outgoing to incoming yield stresses considered. The contact area is calculated using a different plugin called the Zouhar Contact Model discussed below. This leaves the inverse forming efficiency which is calculated using Eq.12.

$$Q_F = \frac{\sigma_R}{k_{f,m}} + 2 \sqrt{\frac{1-|\epsilon|}{|\epsilon|}} \tan^{-1} \left(\sqrt{\frac{|\epsilon|}{1-|\epsilon|}} \right) - 1 + \sqrt{\frac{R}{h_1}} \sqrt{\frac{1-|\epsilon|}{|\epsilon|}} \log \left(\frac{\sqrt{1-|\epsilon|}}{1-|\epsilon|(1-\beta_n^2)} \right) \quad (12)$$

In Eq.12, the term σ_R is equal to the reverse tension applied, ϵ is equal to the relative draught or reduction of the pass, R is the roll radius, h_1 is the outgoing equivalent height, and β_n is the neutral line angle which is calculated using Eq.13.

$$\beta_n = \sqrt{\frac{1-|\epsilon|}{|\epsilon|}} \tan \left(\frac{1}{2} \sqrt{\frac{h_1}{R}} \left[\frac{\sigma_R - \sigma_V}{k_{f,m}} + \log(1-|\epsilon|) \right] + \frac{1}{2} \tan^{-1} \sqrt{\frac{|\epsilon|}{1-|\epsilon|}} \right) \quad (13)$$

In this equation all terms have the same meaning as Eq.12 with the additional term σ_V representing the forward tension applied.

Zouhar Contact Model

This plugin is used to calculate the contact area between the workpiece and the rollers. The base method of calculating the contact area is by using the trapezoidal rule and taking the incoming contact width, outgoing width, and contact length as inputs. This is a very simple method of calculating the area but not very accurate so the Zouhar model uses empirical

correction coefficients for certain geometry combinations to obtain more accurate values. This corrected model is given by Eq.14.

$$A_d = \left[b_1 C_2 + \frac{1}{2} (b_1 + b C_1) (1 - C_2) \right] C_3 L_d \quad (14)$$

In this equation, the term b represents the initial contact width, b_1 represents the outgoing width, L_d represents the contact length, and the terms C_{1-3} represent the correction coefficients which are given in table 3. For geometries not specified in this table, the values default to 1, 0, 1 for C_1 , C_2 , and C_3 respectively which reverts back to a standard trapezoidal rule.

Table 3: Zouhar correction coefficients for given profile and groove geometries

In Profile	Out Profile	C_1	C_2	C_3
Diamond	Diamond	1	0.3	1
Diamond	Square	1	0.28	1
Square	Diamond	1	0.28	1
Oval	Square	1	0.1	1
Square	Oval	0.82	0.2	1.02
Round	Oval	0.45	0.18	1
Oval	Round	1	0	1

3. Methodology

3.1. Model Selection

In this study, PyRoLL is primarily tested against finite element analysis simulations run in the software, Abaqus. This allows an assessment of PyRoLL's performance against the industry standard and conclusions to be made on what applications, if any, it may be comparable or preferable to use over FEA simulations. The primary outputs that are considered are the output geometry of the workpiece, the vertical forces on the roller and the time efficiency of the simulation itself.

On top of the comparisons between simulation methods, comparisons are made against experimental data where possible because ultimately, PyRoLL is attempting to predict the results of a real world rolling process, not replicate the predictions of FEA simulations. These experimental results not only give an additional benchmark to compare PyRoLL against but they also allow validation of the FEA simulations. There may be discrepancies between the FEA predictions and real world outcomes, and using experimental data allows PyRoLL to be tested as an alternative to finite element methods rather than just a copycat aiming for identical results. The discrepancies can be measured as not only a difference between PyRoLL and FEA but as errors in both methods against the correct result.

It is important that before creating models and running simulations that the process is clearly defined so that there are no errors originating from the input conditions. The model selections below define all the conditions that are to be identical between both PyRoLL and FEA simulations.

3.1.1. Flat Rolling Model

The first model selected is a flat rolling process based on a Bekaert tool used to quickly predict the outcome of flat rolling processes. This tool is based on experimental data and is used in industrial applications so it provides reliable output data to compare against PyRoLL. As the priority for this project is shape rolling, this model provides a good starting point to verify PyRoLL against a process with less parameters to consider while also giving more time to become familiar with the software.

Unfortunately, due to commercial sensitivity, access is not available to the Bekaert tool but instead data is available from two simulations run using this tool which are replicated as closely as possible in PyRoLL. Access to only the input and output data limits how much any

discrepancies can be explained but it is a good starting point to compare PyRoLL against a dedicated tool that has been developed specifically for industry with results that are representative of real world processes.

Both flat rolling models consist of an incoming round bar being passed through three sets of rolls with progressively smaller roll gaps. The difference between both models is the diameter of the incoming material and consequentially, the gap between the flat rollers. Table 4 below shows the parameters of both models with the designation of 3 mm flat wire and 14 mm flat wire which represent the height of the wire after the final pass.

Table 4: Parameters of flat wire models based on Bekaert reference tool

Parameter	3 mm Flat Wire	14 mm Flat Wire
Incoming Wire Diameter	10 mm	20 mm
Roll Gap (Pass 1)	6 mm	16 mm
Roll Gap (Pass 2)	4 mm	15 mm
Roll Gap (Pass 3)	3 mm	14 mm
Material Yield Strength	900 MPa	900 MPa

The output data of interest which will be compared to the PyRoLL predictions are the vertical force on the rolls and the width of the outgoing workpiece. This is measured after all three rolling passes on both the 3 mm flat wire and the 14 mm flat wire. This gives a total of 12 data points which are compared against PyRoLL, given in table 5 below.

Table 5: Expected width and force values for flat wire simulations based on reference tool

Roll Pass	Outgoing Width	Roll Force
3 mm Flat Wire – Pass 1	13.002 mm	225.380 kN
3 mm Flat Wire – Pass 2	15.627 mm	445.119 kN
3 mm Flat Wire – Pass 3	17.259 mm	383.438 kN
14 mm Flat Wire – Pass 1	21.777 mm	305.657 kN
14 mm Flat Wire – Pass 2	22.297 mm	413.987 kN
14 mm Flat Wire – Pass 3	22.914 mm	432.782 kN

One concern with this model is that the tool accounts for strain hardening but does not give the relationship used to model this process. A strain hardening relationship is developed that aims to closely represent the strain hardening in the Bekaert reference tool and is discussed further in section 3.2.1. below on PyRoLL model development for this flat rolling simulation.

3.1.2. Round-Oval Model

The first shape rolling model is based on a research paper which aims to assess a semi-analytical model developed by the authors (Lee and Kim, 2001). The model developed in this paper is not used as a reference but rather the experimental data that was used to validate the new model. This is chosen as both the simulations run in PyRoLL and Abaqus are aiming to predict the outcome of real world rolling processes, not copy the results from other models which aim to predict the same thing. Below is described the experimental model that the PyRoLL and Abaqus models attempt to replicate.

The reference model is a round-oval roll pass, meaning a metal rod with a circular cross section is passed through rollers with oval shaped grooves to produce a deformed metal rod with an oval shaped cross section. The incoming workpiece consists of cylindrical rods of plain carbon steel with a 0.1% carbon content. The rods are 60 mm in diameter and 300 mm in length. The rolls are said to have a diameter of 310 mm and during the rolling process they are set to a rolling speed of 34 rpm. The rolls are not flat as in the previous model but instead they have grooves inscribed to produce the oval cross section. The shape of the grooves is shown in Fig. 6 where W_f is the usable groove width, R_1 is the radius of curvature, G is the roll gap, and H_p is the height of the pass and all dimensions are in mm.

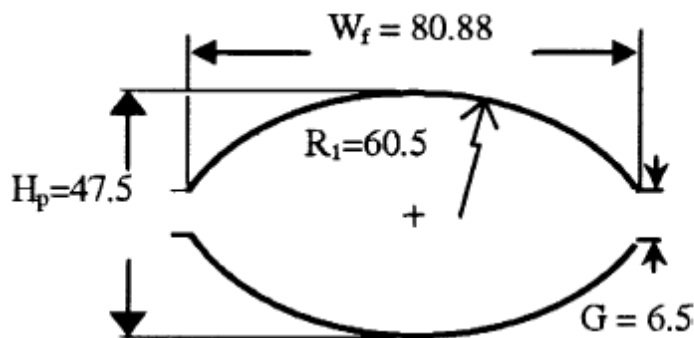


Figure 6: Groove shape from experimental round-oval roll pass

The outgoing cross section from both the model prediction and experimental results are shown in Fig. 7 along with values for the cross sectional area. The maximum width of the experimental cross section is 68 mm.

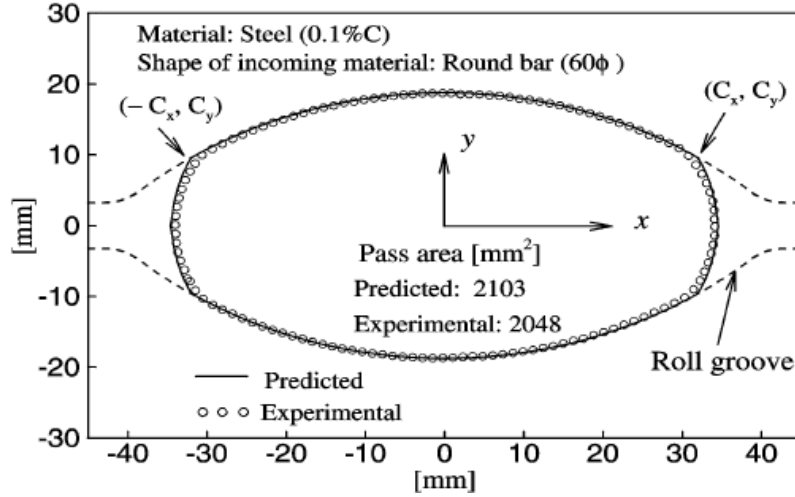


Figure 7: Experimental outgoing workpiece geometry from round-oval pass

The roller forces require more calculations as they are not explicitly given but instead, the model predictions are given as a ratio to the experimental results, presumably to maintain commercial sensitivity while still providing an assessment on the usefulness of the model. This means in order to obtain the experimental forces, the models predicted forces must be calculated as described in the research paper and then using the ratio of predicted to measured force, the experimental roll force can be estimated. Fig.8 shows the ratios of predictions under various conditions when compared to the measured experimental results with the leftmost column, representing rolling at 800°C, being the conditions which are closely replicated in PyRoll and Abaqus.

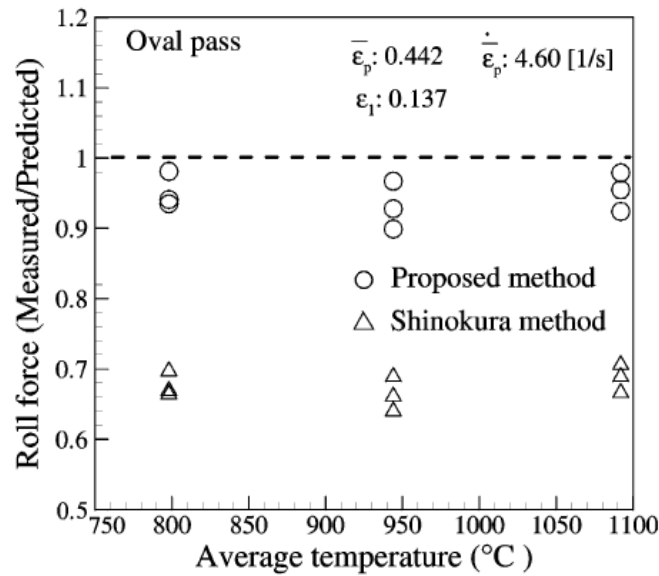


Figure 8: Correlation between experimental roll force measurements and predictions from model proposed by Lee & Kim

The calculations to determine the roll forces are explained in more detail in Appendix A. After these calculations, the experimental values taken for reference are an outgoing width of approximately 68 mm and a vertical roll force in the range 700-735 kN depending on the error of the model proposed in the research.

3.1.3. Square-Diamond Model

The third model is a square-diamond geometry roll pass based on research investigating the temperature, force, and torque during hot bar rolling processes (Said et al., 1999). This research paper specifies four rolling passes but the pass of interest is the second pass, consisting of a square shaped cross section, oriented with its corner facing upwards, being passed through a diamond shaped groove which will reduce the height and increase the corner angle facing upwards.

The incoming geometry is a square section of material with side length 14.58 mm. The groove shape is based on a diamond with a side length of 13.77 mm and a corner angle of 130°. Based on the side length and angles given, the corner to corner height of this groove shape is 11.85 mm and the corner to corner width is 24.85 mm. The rolls are defined as having a 135 mm diameter and a roll speed of 10 rpm.

This research, similarly to the previous paper, carried out rolling procedures under different conditions to investigate how the results change. The force values measured at different temperatures are shown in Fig.9. The third highest force value from this graph is taken as the reference for the square to diamond models with a force value of approximately 95 kN at a temperature of approximately 750°C. This paper quotes Shida's Equation as does the previous study for determining the yield stress of the material. The steel used in this study has a 0.19% carbon content which makes the yield stress less definitive as the diagram doesn't specify yield stress for steel of this carbon content. Linear interpolation is used at 750°C between the lines representing steel with 0.1% and 0.5% carbon content which gives an approximate yield stress of 250 MPa. It is also assumed that there is no strain hardening again as this roll pass is once again carried out well above the recrystallisation temperature of steel.

Although the reference force measurement doesn't require additional calculations for this model, there is no available data on the outgoing cross section of the workpiece and as a result cannot be compared with the PyRoLL and Abaqus models.

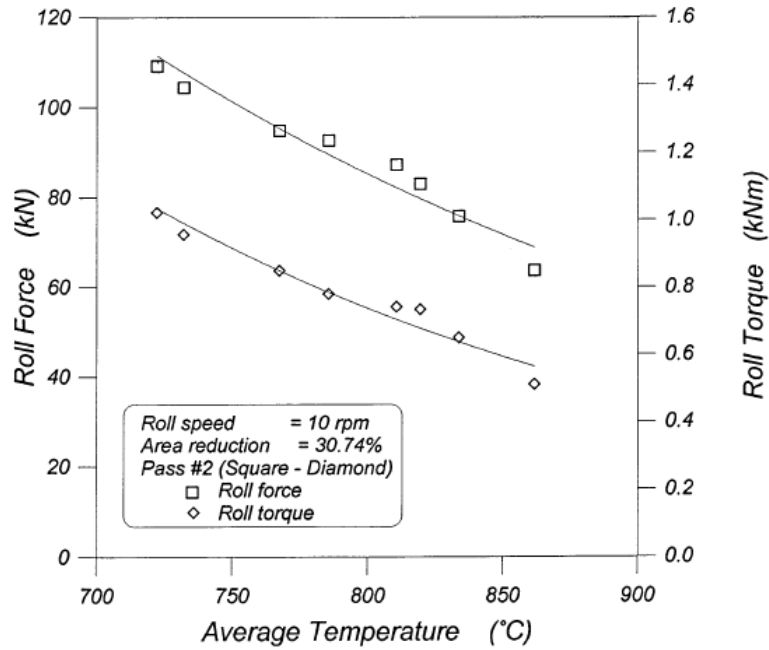


Figure 9: Experimental roll force values for square-diamond roll pass model

3.1.4. Constricted Box Model

The previous models consisted of a flat rolling process and two of the more common shape rolling processes but PyRoIL has the ability to simulate more complex shapes. Unfortunately, there is a limited amount of literature which provide both suitable parameters to replicate a simulation and also experimental data to compare the results of the simulation with. For this reason, a more complex geometry has been developed from scratch for comparison between PyRoIL and Abaqus but without the benefit of experimental data to validate the results.

This model is based on the constricted box geometry available in PyRoIL as it is one of the more complex shapes available to simulate in the programme. As this is developed from scratch, the process of specifying parameters was more of a trial and error process and so is discussed in more detail in the PyRoIL model development section.

3.2. PyRoIL Model Development

3.2.1. Flat Rolling Model

For the development of the flat rolling validation testing, two models are available from the Bekaert tool and used as a reference. The smaller of the two wires (10 mm incoming diameter) is used as a calibration model to iron out any kinks while becoming familiar with the software and noting any potential issues that may cause trouble in later models. After fixing issues with

the first model, changes are made to replicate the larger diameter wire without making any additional changes unless explicitly mentioned in the reference data.

Incoming Workpiece

The incoming workpiece is created using the round profile function. The diameter is defined as 10 mm which is enough to fully define the cross sectional profile.

The yield stress must be defined in order to make predictions based on deformation resistance including the roll force. The yield stress is 900 MPa for the undeformed workpiece as defined in the reference data but instead of a strain hardening relationship, the data only provides the yield stress after each pass. The Freiberg Flow Stress function is used to define the yield stress behaviour. Initially this is set to a constant 900 MPa by setting all hardening coefficients to 0 but can be adjusted to provide more realistic representations of this model.

Additional values are required for the simulation to run but do not have any effect on the outgoing geometry or the roll force so are set to generic values that are representative of the process. The temperature is set to 298.15 K (25°C) which is representative of cold rolling of steel while the density and specific heat capacity are set to 7,850 kg/m³ and 500 J/kg·K respectively which are representative values for steel. These values must be defined for PyRoLL to run a simulation as it takes temperature changes and effects into account when making predictions. These thermal effects aren't seen in cold rolling applications such as the one being replicated in this simulation.

Roller Specification

The three passes in this model use identical flat rolls and as a result, only one set of rolls needs to be created. PyRoLL doesn't provide an option for flat rolls so instead, box groove shapes are used to create an effective flat rolling setup. The groove shape is created with a depth of 1 mm which must be accounted for when defining roll gaps and a sufficiently large ground width. The expected workpiece width after the final pass is 17.3 mm so the ground width of the groove is set to 20 mm to ensure only the flat surface comes into contact with the workpiece. The nominal radius of the rollers is set to 200 mm to replicate the specified diameter of 400 mm given in the reference data.

Roll Pass Creation

Three roll passes are to be simulated for this model and must be created individually. The first input for each pass is the roll which has already been defined above. The primary difference between each pass is the roll gap which gets progressively smaller as the workpiece becomes more deformed. The data specifies roll gaps of 6 mm, 4 mm, and 3 mm but the depth of the box groove must be accounted for when setting the gap, 1 mm on top and bottom rollers. The gaps are therefore set to 4 mm, 2 mm, and 1 mm respectively. The velocity must be defined for each roll pass and is taken from the reference data without issue.

Pass 1 is fully defined after these inputs but not passes 2 and 3. PyRoLL automatically rotates workpieces between passes based on their geometries as is typically done in real world applications. Based on the box shaped grooves used to replicate flat rolls, PyRoLL thinks that a box shaped profile is passing into box shaped grooves and automatically rotates it 90 degrees. This isn't an issue for the first pass which is in the correct position as specified but for passes 2 and 3, the rotation must be specified to be 0 degrees as no rotation takes place in flat rolling.

The cross sections of the roll passes are shown in Fig.10-12.

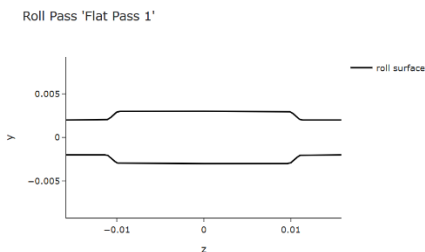


Figure 12: Groove for pass 1 of flat rolling simulation

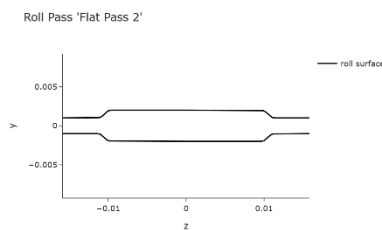


Figure 11: Groove for pass 2 of flat rolling simulation

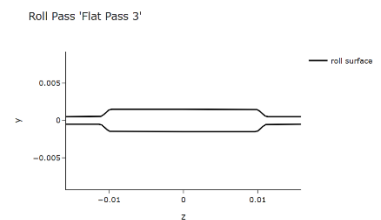


Figure 10: Groove for pass 3 of flat rolling simulation

Running the Simulation

The sequence function is used to create a single roll pass sequence consisting of the three individually defined roll passes. The simulation is run using the solve function for the defined sequence with the previously defined incoming profile as the input for the function. Results and diagrams are automatically produced.

Defining Strain Hardening Relationship

The yield stress is initially set as a constant 900 MPa but the Freiberg Flow Stress function allows more complex relationships to be defined based on strain, strain rate, and temperature.

The reference data shows that the yield stress isn't constant so a strain hardening relationship is implemented. The relationship describing the flow stress in the reference model is not given, instead only the initial yield stress and hardened yield stress after each pass is given. A simplified model is created to replicate these data points as closely as possible. Due to the limited data available, a trial and error approach is used to create a strain hardening relationship that is representative of the reference data. This is done by changing the variable named m2 in PyRoLL to give a general relationship defined by Eq.15 where k_f is the calculated yield stress, A is the undeformed yield stress, ϕ is the strain, and m2 is the coefficient of interest.

$$k_f = A(1 + \phi)^{m2} \quad (15)$$

This relationship doesn't account for strain rate hardening or thermal hardening/softening as strain hardening is the most dominant factor in cold rolling and the limited reference data means extra variables will further complicate things without any actual benefit. After some trial and error the optimum value for m2 was found to be 0.25 and this is used as the material strain hardening relationship in both this model and the larger diameter model.

Larger Wire Model

The purpose of this larger model is to take any lessons learned from the smaller diameter to simulate a model without making changes to obtain results closer to the expected results as any minor adjustments or calibrations should be completed already. This model then is an exact copy of the previous model with changes only made if they are explicitly mentioned in the reference data or required for the simulation to run successfully.

The changes to the previous simulation are an increase in the incoming diameter (from 10 mm to 20 mm), increased roll gaps in line with the new reference roll gaps, updated velocities in line with the new data, and a wider ground width in the groove to accommodate the increased spread from the larger wire (increase from 20 mm to 30 mm). All other inputs remain identical including material properties, rolling temperature, roller radius, etc.

3.2.2. Round-Oval Model

The first shape rolling model to be made in PyRoLL is the round-oval pass. This is developed to replicate the experimental parameters as closely as possible in the paper written by Lee & Kim.

Incoming Workpiece

The incoming workpiece is well defined in the model and can be easily replicated in PyRoll. The workpiece is specified as a cylindrical rod with a 60 mm diameter for the cross section. This is the only necessary input required for the geometry of the workpiece.

The yield stress is set to 200 MPa as predicted based on the calculations and data provided in the literature. No strain hardening relationship is implemented as the experimental rolling took place at 800°C which is above the recrystallisation temperature of steel and hence no strain hardening would take place at this temperature. For this simulation then, a constant yield stress of 200 MPa is assumed.

Roller and Roll Pass Specifications

The groove shape is made to be identical to Fig.6, the diagram on which it is based. The radius of curvature, labelled as r2 in PyRoll is input as 60.5 mm and the usable width is input as 80.88 mm, both values that are specified in the diagram from the literature upon which the groove is based. These two values alone are enough to fully define a circular oval groove.

The actual roll itself is given a nominal radius of 155 mm and a rotational frequency of 0.57 times per second, both calculated easily from the given diameter of the roll and rolling speed in rpm.

The roll pass is given a 6.5 mm roll gap as specified and the previously developed roll is used for the process. The cross section for this pass is seen in Fig.13. The velocity does not need to be defined as the rotational frequency is already specified and rotation is not an issue for a single pass sequence. The pass defined can be directly simulated using the solve function also since it is not part of a sequence consisting of multiple passes.

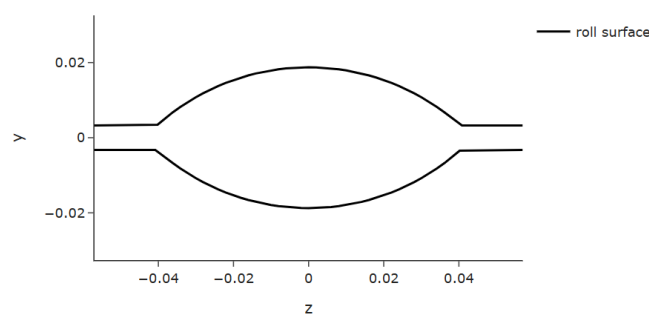


Figure 13: Oval groove shape designed in PyRoll

3.2.3. Square-Diamond Model

The third model developed in PyRoLL is based on the square-diamond model defined in section 3.1.3.

Incoming Workpiece

The incoming workpiece is of different geometry for this model. The square profile function is used which produces a rod/bar with a square cross section oriented with its corners facing upwards and downwards. The square is given sides of length 14.58 mm. PyRoLL also requires corner radii to be defined for square cross sections but these aren't defined in the literature. A corner radius of 1 mm is chosen to avoid sharp edges but not have too great of an effect on the cross sectional shape. This profile is seen in Fig.14.

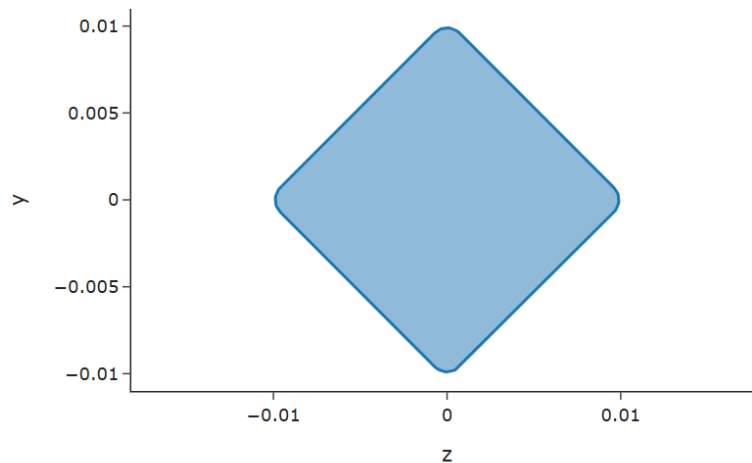


Figure 14: Incoming square cross section in PyRoLL

The yield stress is once again defined as a constant value due to the high temperature at which the experimental rolling took place at (approximately 750°C) meaning that no strain hardening is expected. The yield stress is specified as 250 MPa for this model based on the same data as the round-oval model but with a higher carbon content and slightly lower rolling temperature.

Roller and Roll Pass Specifications

The groove shape for this model is not square but instead diamond shaped. This contains some of the same elements as the square geometry, including four sides of equal length and the orientation with the corners facing upwards and downwards. The primary difference for the diamond geometry is that the inner angles do not equal 90 degrees. For this model, the inner angle at the top corner is equal to 130 degrees. One more input is required to fully define this

diamond shape and the groove depth is defined as 5.425 mm. This is not specified in the model but the corner to corner height of the outgoing diamond is 11.85 mm which can be achieved using two groove depths of 5.425 mm and a roll gap of 1 mm. The corner radius is defined as 1 mm again to match the corner radius of the incoming square profile.

The roll itself is created using the diamond groove defined with a nominal radius of 67.5 mm (taken from the diameter specified in the literature), and a rotational frequency of 0.167 (taken from the rotational speed given in rpm). The roll gap is 1 mm to give a corner to corner height of 11.85 mm with the groove depth specified. The cross section view of the roll pass is shown in Fig.15.

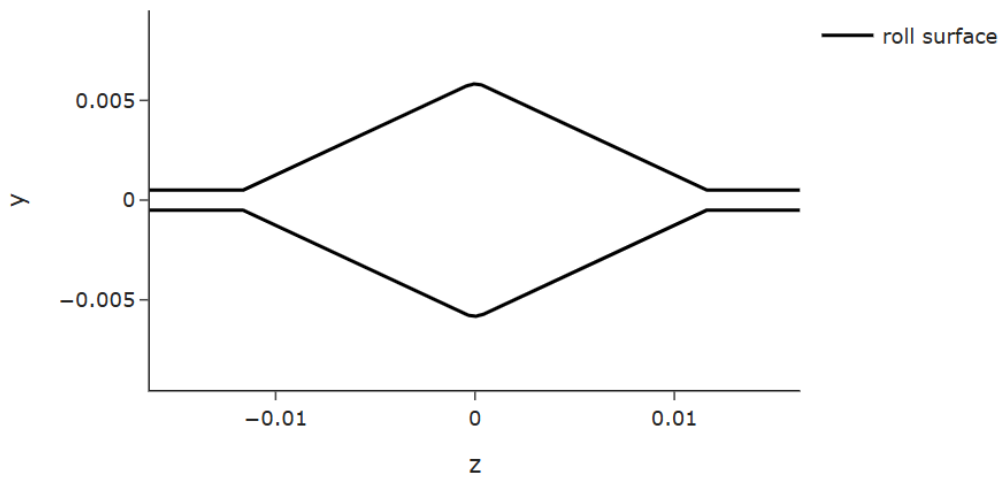


Figure 15: Diamond groove shape designed in PyRoll

3.2.4. Constricted Box Model

This model is slightly different to the previous tests as it is based on a custom model without experimental data so the input parameters aren't supposed to exactly replicate any experimental parameters, but they must still be accurately recreated in the Abaqus model to ensure consistent models and a fair comparison. The model is based on the constricted box geometry available in PyRoll (template shown in Fig.16) with the idea being that a constricted box shape is created and then deformed to increase the indent of the workpiece.

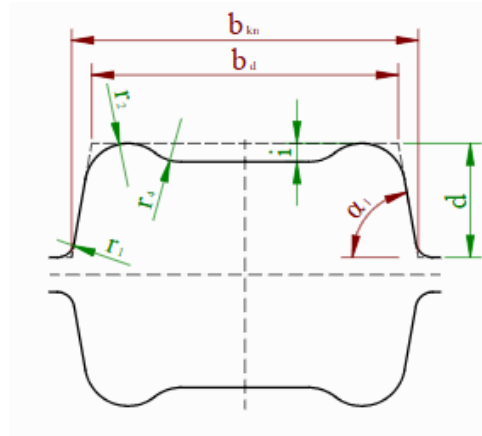


Figure 16: Constricted box template available in PyRoll

Incoming Workpiece

The incoming workpiece is made from a groove shape meaning that the groove must be designed first. The groove is based on the template shown in Fig.16 and as such, all dimensions are referenced from this diagram. The maximum depth, d , is set to 2.1 mm with the indent, i , set to 1.25 mm. These figures will give a significant indent as opposed to just a standard box groove. The corner radii r_2 and r_4 are set to 1 mm and 2.8 mm respectively. These values are sufficiently large to avoid sharp corners which may be difficult to simulate and also difficult to produce in the real world. The radius, r_1 is set to 1 mm to produce an effective sharp corner at the groove edge as no material should be spilling outside of the edge and therefore no material needs to fit around the sharp corner. Finally the usable width, b_{kn} and the ground width, b_d are set to 9 mm and 8.9 mm respectively to create a near vertical surface. This cannot be exactly vertical as PyRoll demands some sort of a slope for this edge otherwise it will return errors.

The actual profile function is very similar to previous models. To define the geometry, instead of inputting the relevant parameters to describe the shape, the previously made shape groove is selected with a filling ratio and roll gap. A filling ratio of 1 is chosen to produce exactly the shape of the groove and a roll gap of 1 mm is defined so that the shape includes enough material for a 1 mm roll gap between the upper and lower rolls. The groove shape and profile produced from it are shown in Fig.17 and Fig.18.

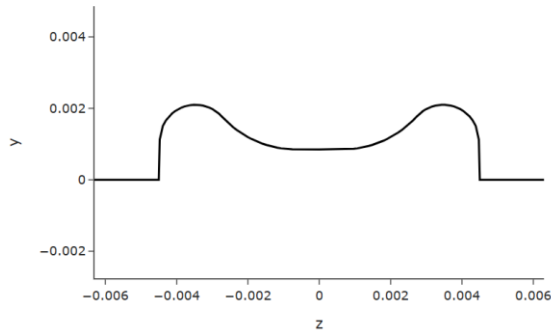


Figure 17: Groove shape used to create incoming constricted box profile in PyRoLL

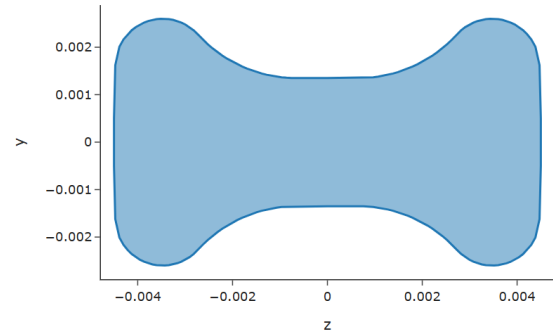


Figure 18: Incoming constricted box profile designed in PyRoLL

The material properties of this model do not need to follow any particular material considering there is no experimental data to replicate. It is only important that the material model is kept consistent between both the PyRoLL and Abaqus models. The material properties are taken from the high strength steel used in the flat rolling validation problem, with an initial yield stress of 900 MPa and a strain hardening relationship described by equation 6 when the coefficient m_2 has a value of 0.25.

Roller and Roll Pass Specification

The groove shape used for the roller is very similar to the groove used to create the incoming profile. A slight reduction in height is obtained by reducing the depth from 2.1 mm to 2 mm and by increasing the indent from 1.25 mm to 1.55 mm. The usable and ground widths are both increased by 1 mm, giving values of 10 mm and 9.9 mm respectively. This both allows space for spreading of the material without overfilling of the groove while also preventing material from putting purely lateral pressure on the sidewalls of the groove. The radii, r_1 , r_2 , and r_4 are all kept at the same values as the groove shape used to create the incoming profile.

The roll is created using the second groove shape with the reduced height and increased width. The radius of the roll is set to 15 mm, much smaller than previous models due to the reduced size of the workpiece and deformations obtained. Rotational frequency is set to 0.318 revolutions per second which is approximately 2 radians per second. Using an even value in radians per second makes speed control easier in Abaqus where the total time and radians are specified rather than actual speeds. The roll gap is set to 1 mm to match the 1 mm roll gap specified when creating the incoming profile, ensuring any deformation occurs only as a result of the change in groove shape rather than rolls moving closer together. Finally, the rotation is set to 0 degrees. Usually, the rotation doesn't need to be changed for the first roll pass but in

this case, the incoming profile is created from a roll groove and is assumed to be coming from this groove and therefore is rotated unless specified otherwise. The roll pass cross section is shown in Fig.19.

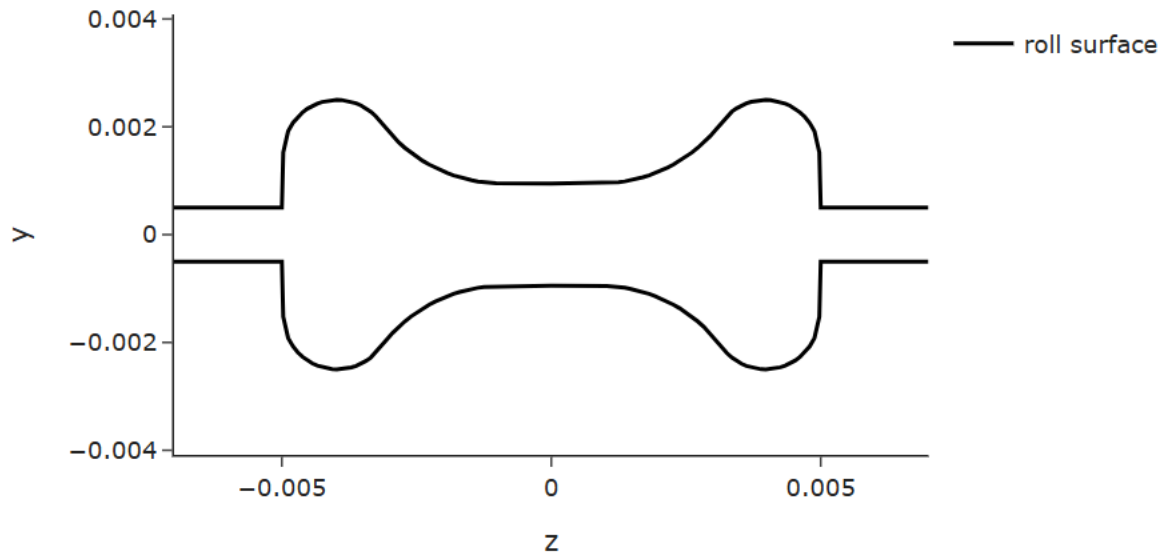


Figure 19: Constricted box groove shape designed in PyRoll

3.3. FEA Model Development

This section outlines how finite element analysis models were developed in Abaqus to compare against the predictions of the PyRoll simulations. No FEA models were developed for the flat rolling validation testing as the purpose of using FEA is to compare PyRoll to an industry standard level of simulation. The Bekaert tool with which the PyRoll model is based is an actual simulation tool used in industry, so serves the same purpose as the FEA models in the shape rolling simulations. The reference results are also the output of the Bekaert simulation so it doesn't make sense to use Abaqus as an industry standard tool for comparison against PyRoll to replicate another industry standard tool.

The models developed in Abaqus are made to replicate the PyRoll models as much as possible. Some values are estimated for the experimental models but the same estimations are used for both the PyRoll and Abaqus development. This ensures consistency between simulations and any discrepancies between PyRoll and Abaqus predictions should be due to the calculation method rather than differences in input data.

3.3.1. Round-Oval Model

Creating the Parts

Firstly, the geometries of the incoming workpiece and the rolls are defined. The simulations in Abaqus make use of symmetry planes whose implementations are described below but for defining part geometries, this means only a smaller portion is defined with the remainder being accounted for by symmetry during simulation.

The incoming workpiece is a cylindrical rod with a diameter of 60 mm. In Abaqus, this is specified to be a deformable, solid part produced by extrusion. A sketch is drawn of a semi-circle with diameter 60 mm to account for the cross section which will be mirrored to create a full circle. The length of the workpiece is not defined in PyRoLL but must be defined in Abaqus to create a 3D part by extrusion. The exact length isn't important but the part must be long enough so that steady state rolling is achieved and both spread and force values are obtained that are representative of a continuous rolling process. An extrusion length of 300 mm is chosen as this is the length of the samples used in the experimental data. This should be suitable therefore to reach steady state rolling while also not being too large so as to significantly increase the duration of the simulation.

The roller is created by drawing a sketch of the cross section, shown in Fig.20, and revolving around the central axis to create a full 3-dimensional roller. The part is defined as a discrete rigid shell to improve efficiency of the simulation. The maximum height of this sketch is 155 mm as given by the roll radius, the total width is 100 mm, enough to house the groove of width 80.88 mm. The groove shape is sketched to the same conditions as the PyRoLL model, a curve of radius 60.5 mm which stops at a distance of 40.44 mm from the centre line at each side to replicate the usable width as defined in the PyRoLL and reference models. The sketch is then revolved 360 degrees around the X-axis, which is marked as a horizontal, yellow, dashed line to create a full roller, shown in Fig.21.

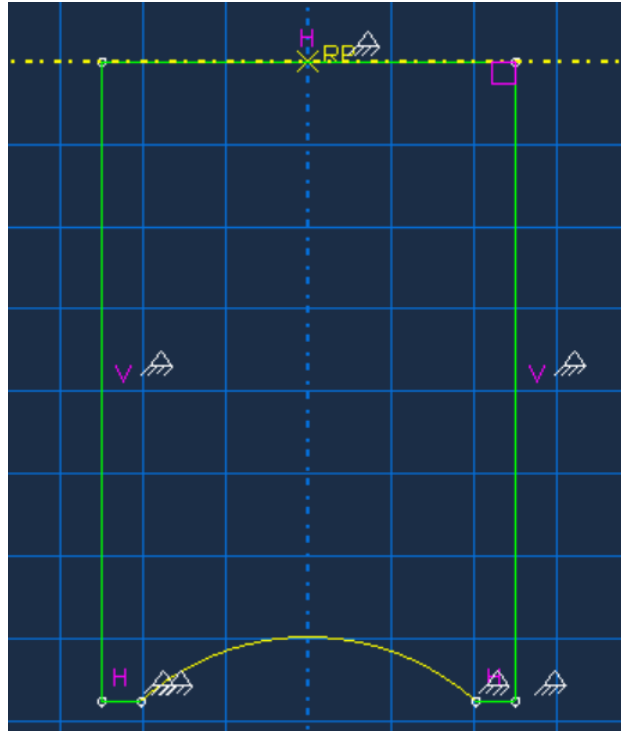


Figure 20: Sketch of roller cross section with oval groove in Abaqus

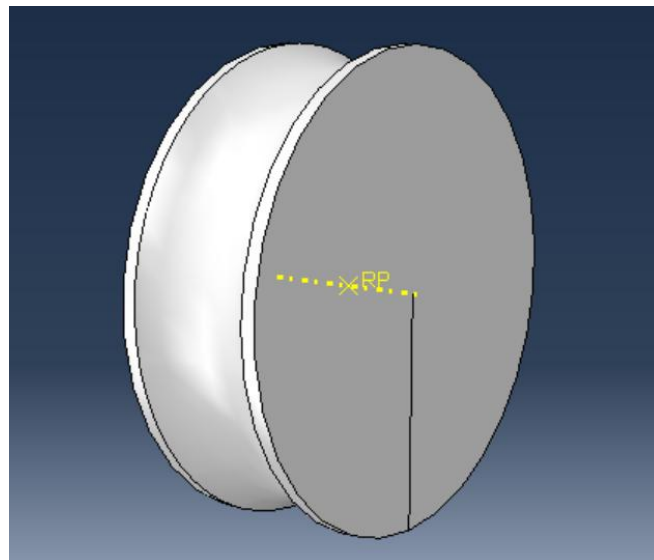


Figure 21: Roller with oval shaped groove in Abaqus

Both parts are meshed once the geometry has been defined with elements of approximately 5 mm in size. This mesh size is small enough to provide relatively accurate outputs while also not being too small that simulations require excessive computing power and can be run on the available equipment. The final parts after meshing are shown in Fig.22 and Fig.23.

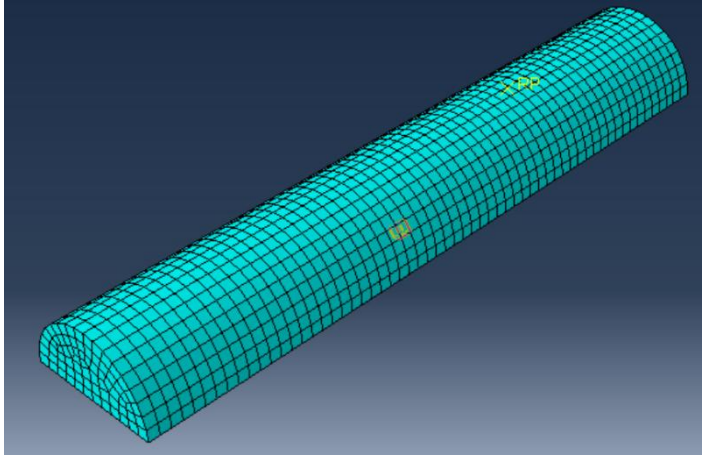


Figure 22: Semi-cylindrical workpiece with 5 mm mesh

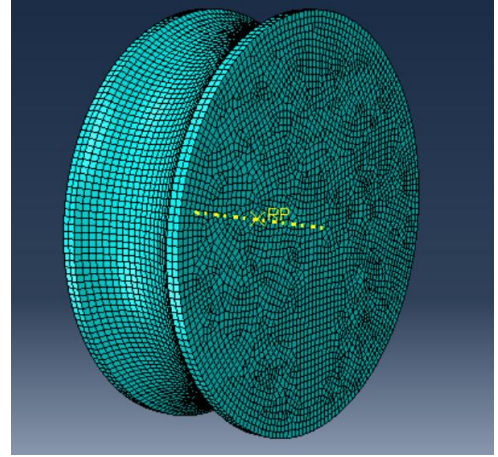


Figure 23: Oval groove shaped roller with 5 mm mesh

The material properties are also required to be defined and a new material is created to represent the steel used in both the experimental testing and PyRoLL model. The material is defined as having a Young's Modulus of 200 GPa, a Poisson's Ratio of 0.3 and a yield stress of 200 MPa as specified in the PyRoLL model with no strain hardening characteristics. These steel properties are applied to the semi-circular workpiece as a steel section. The roller is not given any material properties as it is a rigid body and will not deform during the simulation.

Creating the Simulation

Creating the simulation in Abaqus is much more complex than in PyRoLL. The Abaqus simulation consists of two stages. The first stage is the compression step in which the upper and lower rollers move closer together until the specified roll gap is achieved, compressing the workpiece in the process. The second stage is the rotation step in which the actual rolling takes place. Once the rollers are the correct distance apart, they begin rotating to pull the workpiece through the roll gap and deforming it into the desired shape, eventually reaching steady state rolling which is directly simulated in PyRoLL.

Before creating the Abaqus simulation parameters, the initial assembly must be created. A single instance of the roller and half workpiece are inserted into the assembly and translated to the correct location with respect to the symmetry plane that will mirror the bottom half of the simulation. The centre point of the roller is located on this symmetry plane at first but the roller is translated in the vertical direction by 173.25 mm. This value consists of a 155 mm translation (roll radius) to put the edge of the roller on the symmetry plane, a 3.25 mm translation (half the roll gap) to achieve the correct distance between the defined roller and the mirrored lower roll,

and an additional 15 mm translation which ensures no contact between the parts before the simulation starts. The semi-cylindrical workpiece has its flat surface located on the symmetry plane as intended but is translated by 50 mm in the direction of rolling to ensure that no backwards slip occurs during the compression step. The assembly before simulation is shown in Fig.24 and Fig.25.

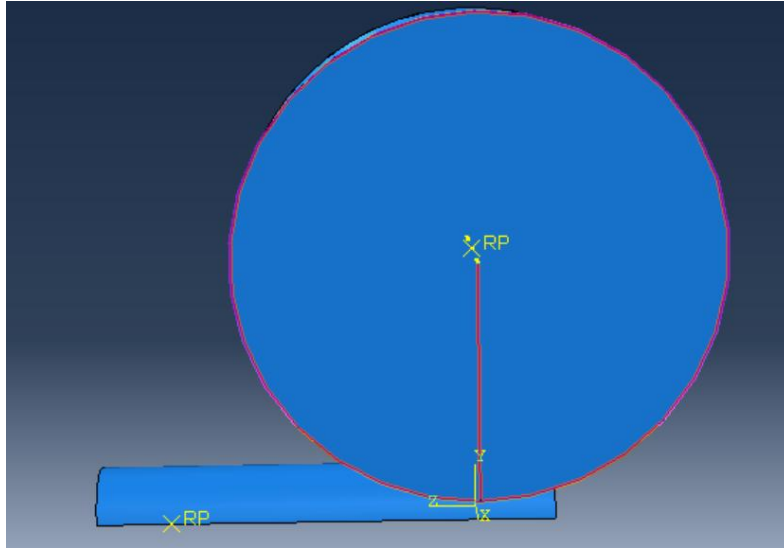


Figure 24: Side profile of round-oval model assembly in Abaqus

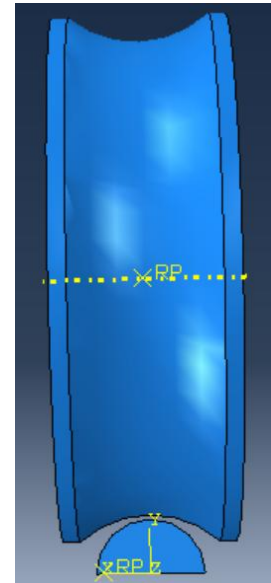


Figure 25: End view of round-oval model assembly

Two steps are created, named compression and rotation which are then used to define boundary conditions and simulation parameters. A surface contact interaction is created for the groove surface and outer surface of the workpiece (Fig.26). This contact interaction also defines a coefficient of friction of 0.3.

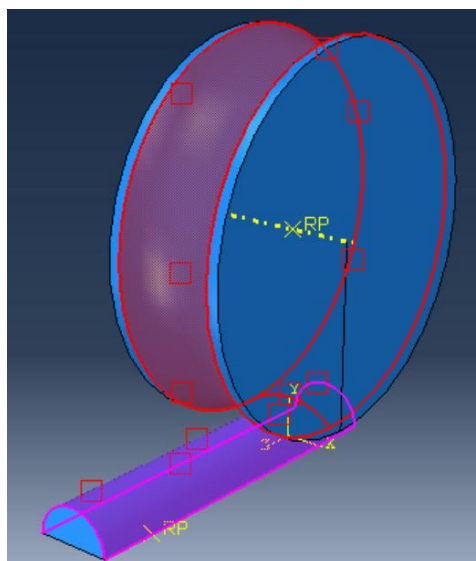


Figure 26: Surface contact interaction of round-oval model in Abaqus

Once these parameters are defined, the actual simulation conditions are created through boundary conditions. A reference point is created at the centre of the roller, labelled RP in Fig.26 through which the actual rolling conditions are applied. During the compression step, a downward, vertical displacement of 15 mm is specified for the reference point to create the desired 6.5 mm roll gap. Displacements in the lateral directions (both parallel and perpendicular to the rolling direction) and all rotations of the reference point are set to 0 for this step to ensure that no unintended movements occur.

For the rotation step, this boundary condition is maintained to avoid unwanted movements of the roller. An additional rotation boundary condition is specified with a rotation of 2 radians about the axis perpendicular to the direction of rolling (marked in Fig.26 by the yellow, dashed line) to perform the actual rolling deformation. To avoid conflict, the previous boundary condition is edited so that it does not specify rotation in this direction. The rotation of 2 radians is large enough to achieve steady state rolling while also not being so large as to go beyond the end of the workpiece and waste simulation time where no deformation occurs.

A symmetry boundary condition is applied in the y-direction (vertical) with the flat surface on the bottom of the semi-cylindrical workpiece acting as the symmetry plane. This ensures that although only half a model is being simulated, the results should replicate the outcome of a simulation of the entire model with full workpiece and two rollers.

The final boundary condition is applied to the two semi-circular end faces of the workpiece specifying that they can only move in the direction of rolling. The purpose of this is to eliminate any unwanted twisting or distortion of the workpiece by ensuring the faces cannot move in the lateral or vertical directions and cannot be rotated in any direction.

Creating Data

PyRoLL runs simulations and automatically calculates a wide range of outputs such as the outgoing geometry and the roll force but these aren't calculated in Abaqus unless specified before running the simulation. The data in this Abaqus simulation is recorded using history output requests.

To record the roller forces, a history output request for the reaction force RF2 (reaction force in the vertical direction) is set using the roller reference point as the domain. This records the vertical forces that the reference point is subjected to throughout the entire rolling process, producing a time dependant graph from which the steady state rolling force can be obtained.

To record the maximum width of the outgoing workpiece, a number of sets are created consisting of individual nodes at the outermost edge of the undeformed workpiece. One of these nodes is marked in Fig.24-26 by a reference point on the workpiece labelled RP. A history output is created for each of these sets for the displacement U1 (displacement in direction perpendicular to rolling). This produces a time dependant graph for each of these selected nodes in the lateral direction for the entire rolling process.

Due to the symmetry and end face boundary conditions, the lateral displacement of these nodes should be entirely confined to the deformation zone, meaning once the node has passed through the rollers a discrete displacement should be obtained. Some nodes are rolled before steady state rolling is achieved and some nodes don't fully pass through the rollers so the use of multiple displacement nodes provides enough datapoints to determine what the displacement value is under steady state rolling. Unlike the roll forces, further calculations are required to determine the maximum outgoing width as opposed to reading values directly from the graph. The displacement measured is the increase in width on one side of the workpiece, so to calculate the maximum outgoing width, the displacement is doubled to account for both sides of the workpiece and the result added to the incoming width.

Once all the simulation parameters are specified, a job is created and submitted to run the simulation.

3.3.2. Square-Diamond Model

Creating the Parts

Similarly to the round-oval model, the first step is to create the undeformed workpiece and roller geometries. In addition to using different shaped geometries to the previous model, this model also makes use of an additional symmetry plane. This means not only is the top half of the model mirrored onto the bottom half but also the right side is mirrored onto the left, further reducing the number of elements to be simulated and increasing efficiency.

The incoming workpiece consists of a square shaped rod with side length of 14.58 mm and a corner radius of 1 mm. A quarter of this cross section is sketched as shown in Fig.27 to be mirrored twice during simulation, producing a full square shaped cross section. This sketch is extruded a length of 200 mm to provide a suitable length to achieve steady state rolling. The extruded piece is shown in Fig.28.

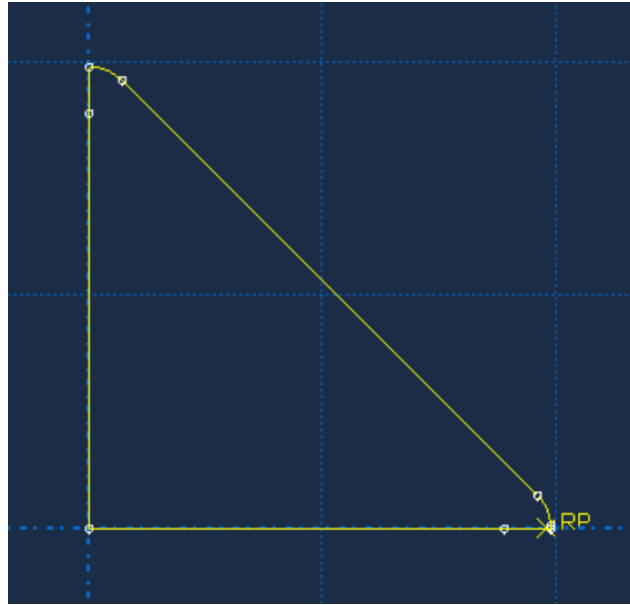


Figure 27: Sketch for quarter square shaped workpiece in Abaqus

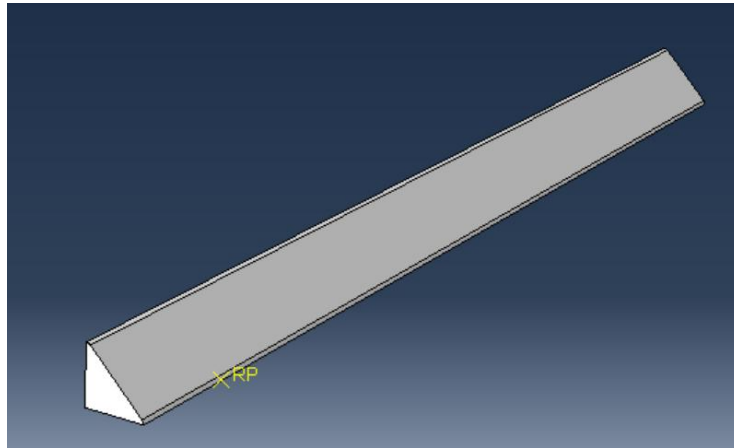


Figure 28: Quarter square shaped workpiece in Abaqus

The roller is created using the same method as the round-oval roller, creating a sketch of the cross section and revolving around the central axis with the difference being that only half of the roller is sketched to allow for symmetry to mirror the other half. The sketch is shown in Fig.29 with the height equal to the roll radius of 67.5 mm, the total width equal to 50 mm to accommodate the entire groove width, and the groove shape designed to exactly replicate the geometry specified in the model selection and PyRoLL model once mirrored along the centreline. The part is then produced by revolving around the central axis to produce the rigid shell shown in Fig.30 with the flat surface shown being coincident with the symmetry plane during the actual simulation.

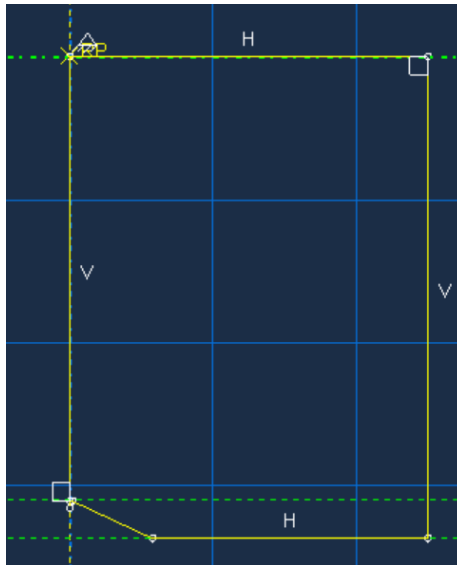


Figure 29: Sketch for half roller with diamond groove shape in Abaqus

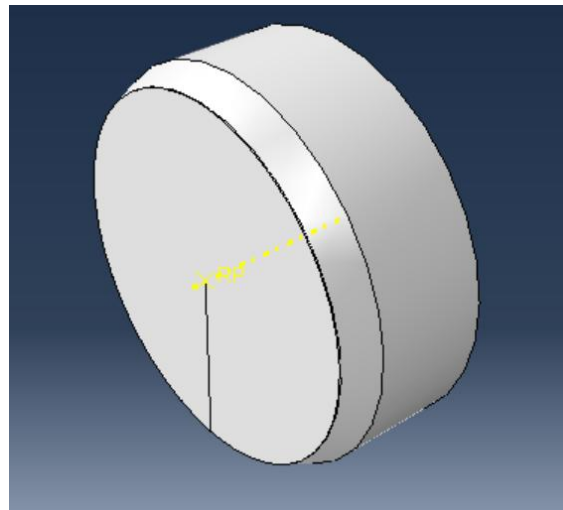


Figure 30: Half roller with diamond groove shape in Abaqus

Meshes are created for both parts with approximate size of 5 mm for the roller and approximate size of 2 mm for the workpiece. The smaller mesh size is selected due to the smaller workpiece size which will undergo smaller deformations and require smaller elements. The smaller mesh decrease in efficiency caused by smaller mesh sizes is not felt as much due to the fact this model is already smaller than the previous model and the extra symmetry plane means only a quarter of the model is simulated compared to a half, greatly reducing the number of elements and computational time. The meshed parts are shown in Fig.31 and Fig.32.

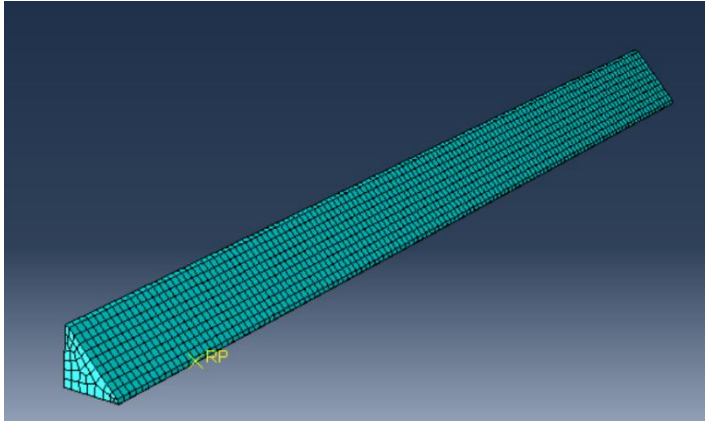


Figure 31: Quarter square shaped workpiece with 2 mm mesh in Abaqus

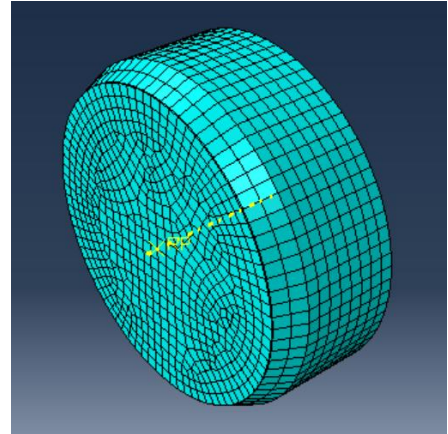


Figure 32: Diamond groove shaped roller with 5 mm mesh in Abaqus

The material properties are defined with a yield stress of 250 MPa and all other values are identical to the steel created for the round-oval model. A steel section is created and applied to only the workpiece because the roller is rigid and does not require material properties.

Creating the Simulation

The simulation consists of two steps like the round-oval model, a compression step in which the rollers move closer together to achieve the correct roll gap followed by a rotation step where they rotate to pull the workpiece through the roll gap and produce the desired deformation.

The initial assembly is created and set up in a similar manner to the round-oval model with one instance each of the partial workpiece and partial roller. The roller is translated vertically by a distance of 78 mm, accounting for the 67.5 mm roll radius, 0.5 mm for half the roll gap and an additional 10 mm to allow room for the compression step. The bar is translated 30 mm in the roll direction to prevent backwards slip during the compression step. The assembly before simulation is shown in Fig.33. Once the assembly is created, the compression and rotation steps can be created.

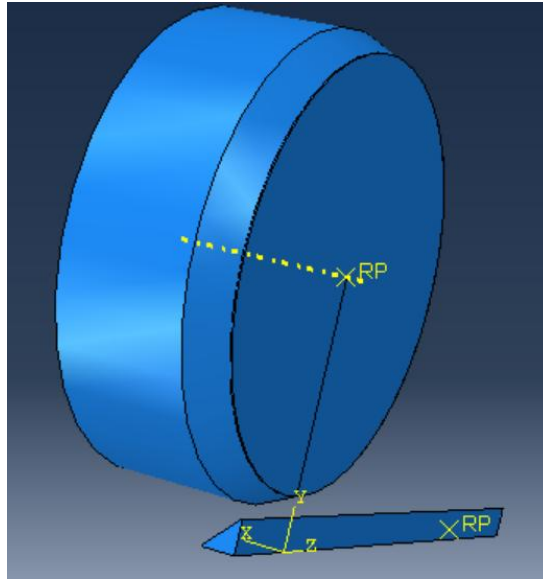


Figure 33: Square-diamond model assembly in Abaqus

A surface contact interaction is created between the roller and workpiece surfaces (Fig.34 and Fig.35) with a coefficient of friction of 0.3.

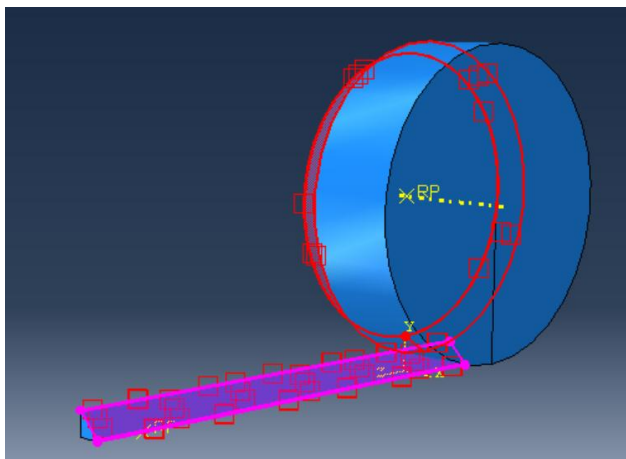


Figure 34: Surface contact interaction for square-diamond model in Abaqus (1/2)

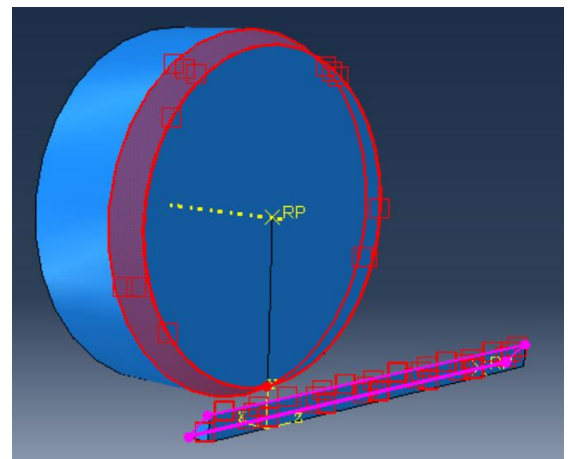


Figure 35: Surface contact interaction for square-diamond model in Abaqus (2/2)

The boundary conditions created are very similar to the round-oval model. A roller reference point is created at the centre of the roller which is used for the boundary conditions which actually perform the rolling deformations. A compression boundary condition is created which specifies a downward translation for this reference point of 10 mm while preventing any other translations or rotations. A rotation boundary condition is then created for the rotation step which specifies a rotation of 2 radians around the axis marked in Fig.33 by the yellow, dashed line. Once again, the compression boundary condition is edited to not specify rotation about this axis for the rotation step to avoid conflict between different boundary conditions.

The symmetry boundary conditions differ in this model compared to the round-oval model as only a quarter of the model is defined rather than a half. A y-symmetry boundary condition is defined using the flat surface on the bottom of the quarter workpiece as the symmetry plane. An x-symmetry condition is also created using the flat surface at the inside of the roller which is visible in Fig.33. There is no boundary condition created for the edge faces of the workpiece in this model as the double symmetry prevents any possible distortion or twisting of the workpiece and the faces can only move in a straight line parallel to the z-axis.

Creating Data

The data is recorded in the same manner as the round-oval model by using history output requests. The roller forces are recorded using a history output for the reaction force RF2 (reaction force in the vertical direction) with the roller reference point used as the domain, producing a time dependant graph from which steady state rolling can be identified and a roll force obtained.

The maximum width of the outgoing workpiece is recorded by once again creating multiple sets consisting of individual nodes at the outermost edge of the undeformed workpiece (one example can be seen in Fig.31 labelled RP) and creating history output requests for the U1 displacement of these nodes (displacement in the direction perpendicular to rolling). This history output records the increase in width for one side of the workpiece which can be doubled to obtain the full increase in width and then added to the original width to determine the maximum outgoing width of the workpiece. Once again, multiple displacement nodes are specified to compare with one another and confirm that the final obtained width is in fact the steady state outgoing width.

Once these history output requests are finalised, a job is created and submitted to run the simulation. For this model, a mesh sensitivity analysis is complete with 1.5 mm and 1 mm mesh sizes for the part and all other aspects of the simulation remaining unchanged.

3.3.3. Constricted Box Model

Creating the Parts

The constricted box FEA model uses two planes of symmetry so only a quarter of the workpiece and half of the roller are defined with the remainder of both geometries mirrored by the symmetry planes.

The cross section for incoming workpiece is sketched to replicate exactly the cross section defined in the PyRoLL model when both symmetry planes are applied (seen in Fig.36). Accounted for are all the parameters required to define a constricted box groove geometry in PyRoLL and also the roll gap of 1 mm used in the PyRoLL model to define the incoming geometry. The sketch is extruded to a length of 100 mm to produce the part used in the simulation (shown in Fig.37), shorter than the previous simulation to once again improve efficiency of simulations without affecting its ability to achieve steady state rolling due to the smaller cross section.

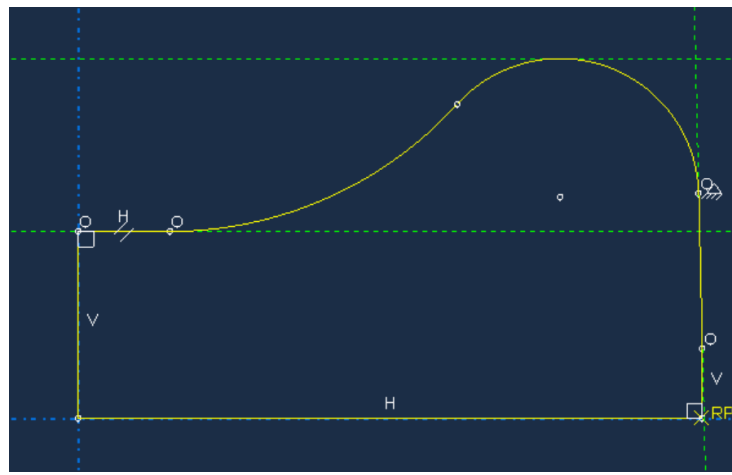


Figure 36: Sketch for quarter constricted box shaped workpiece in Abaqus

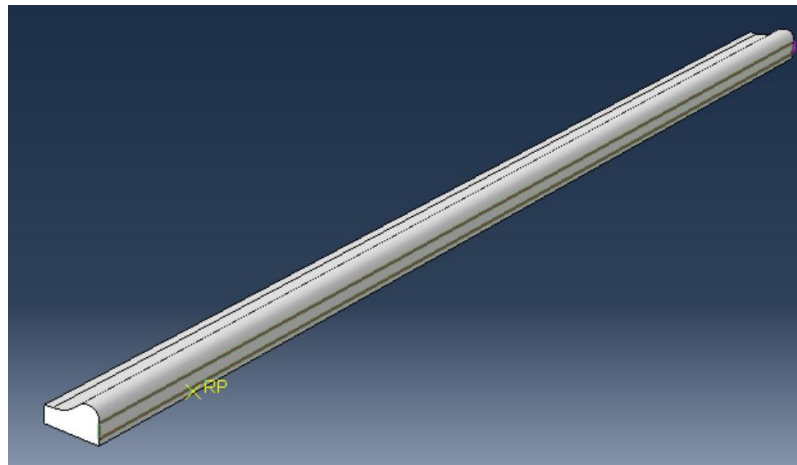


Figure 37: Quarter constricted box shape workpiece in Abaqus

The roller is sketched to exactly replicate the groove shape and roll defined in the PyRoLL model after symmetry is applied (shown in Fig.38). The inputs include the parameters required to define the constricted box groove shape in PyRoLL, a height of 15 mm to replicate the roll radius, and a width of 10 mm to accommodate the entire groove width. Like the previous

models, this sketch is revolved around a centreline to make a half roller (shown in Fig.39 and Fig.40) which is completed once symmetry is applied.

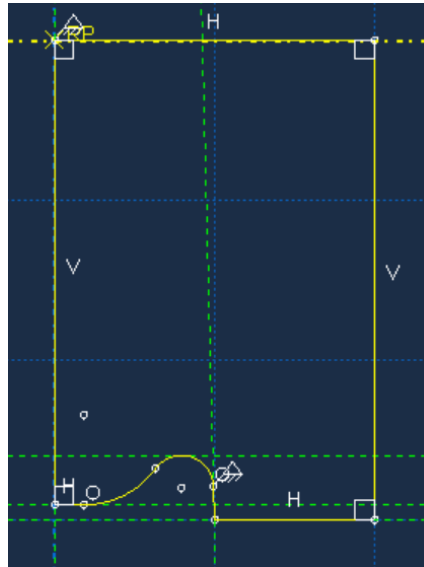


Figure 38: Sketch for half roller with constricted box groove shape in Abaqus

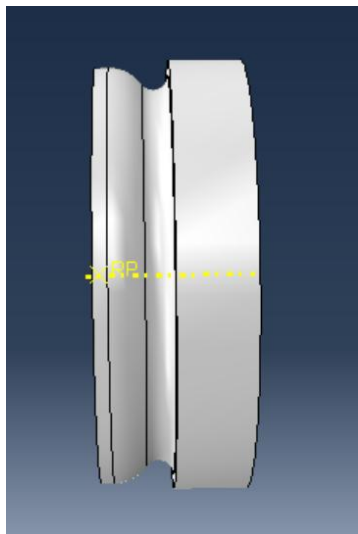


Figure 39: End view of constricted box groove shaped roller in Abaqus

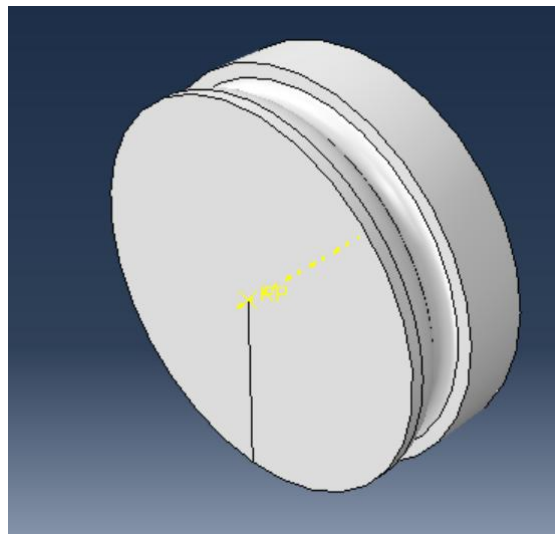


Figure 40: Isometric view of constricted box groove shaped roller in Abaqus

Meshes are created for both parts with approximate size of 1 mm for the roller and approximate size of 0.5 mm for the workpiece. A significantly smaller mesh size is chosen compared to previous models due to the smaller workpiece and roll geometry. This smaller geometry both allows smaller meshes to be used without creating so many elements that the simulation time becomes excessively large and also necessitates a smaller mesh size to accurately simulate the smaller geometries and deformations taking place. Both meshed parts are shown in Fig.41 and Fig.42.

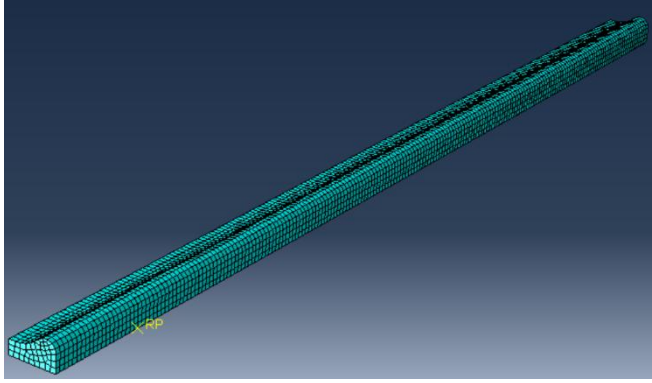


Figure 42: Quarter constricted box shaped workpiece with 0.5 mm mesh in Abaqus

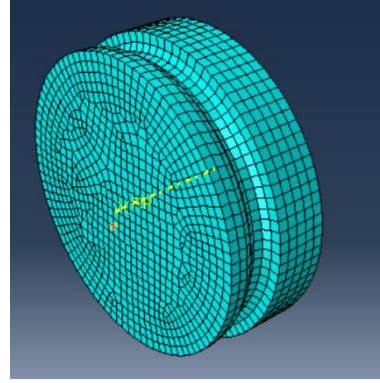


Figure 41: Half constricted box groove shaped roller with 1 mm mesh in Abaqus

The material must be created in order to simulate the deformation of the workpiece. The elastic properties are copied from the previous models, consisting of a Young's Modulus of 200 GPa and a Poisson's Ratio of 0.3. A strain hardening relationship is defined in the PyRoLL model given by Eq.15 where the initial yield stress, A , is 900 MPa and the strain hardening coefficient, m_2 , is 0.25. Replicating this relationship in Abaqus is done by inputting the yield stress values at different strain conditions and using linear interpolation to fill in the gaps. Using enough data points with this method effectively replicates the strain hardening relationship. The values input into Abaqus to replicate this are given in Table 6. Once the material is created, a steel section is defined and applied to the workpiece but not the roller.

Table 6: Yield stress values used for constricted box model in Abaqus based on strain

Strain	Yield Stress (MPa)
0.00	900.000
0.05	911.045
0.10	921.702
0.15	932.002
0.20	941.972
0.25	951.634
0.30	961.011
0.35	970.121
0.40	978.982
0.45	987.608
0.50	996.014
0.55	1004.212
0.60	1012.214

0.65	1020.031
0.70	1027.673
0.75	1035.147
0.80	1042.463
0.85	1049.628
0.90	1056.649
0.95	1063.533
1.00	1070.286

Creating the Simulation

As in the previous models, the simulation consists of two steps, compression and rotation which move the rollers together to obtain the correct roll gap and rotate to create the desired deformation respectively.

The assembly is created and instances translated into correct positions. The roller is translated vertically by a distance of 17.5 mm which accounts for the roll radius of 15 mm, half the roll gap which is equal to 0.5 mm, and an extra 2 mm to allow space for compression. The workpiece is translated 5 mm in the direction of rolling to prevent backwards slip during compression. Fig.43 and Fig.44 show the assembly with instance positioning before the simulation is carried out.

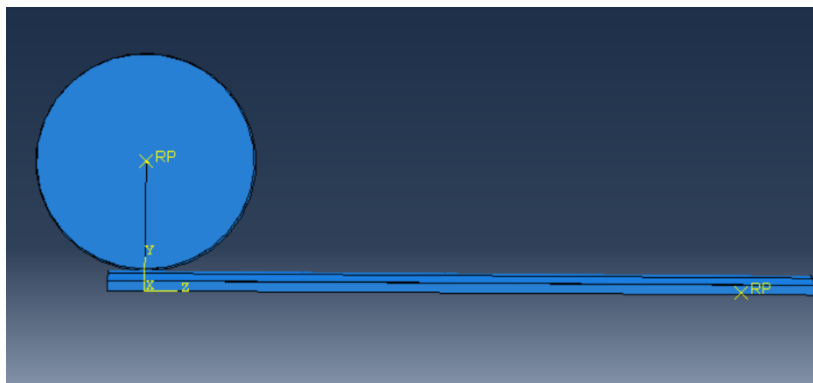


Figure 44: Side view of constricted box model assembly in Abaqus

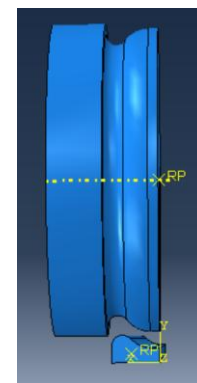


Figure 43: End view of constricted box model assembly in Abaqus

Once the assembly is created, the compression and rotation steps are created. A surface contact interaction is created between the roller and workpiece surfaces (Fig.45) with a defined friction coefficient of 0.3.

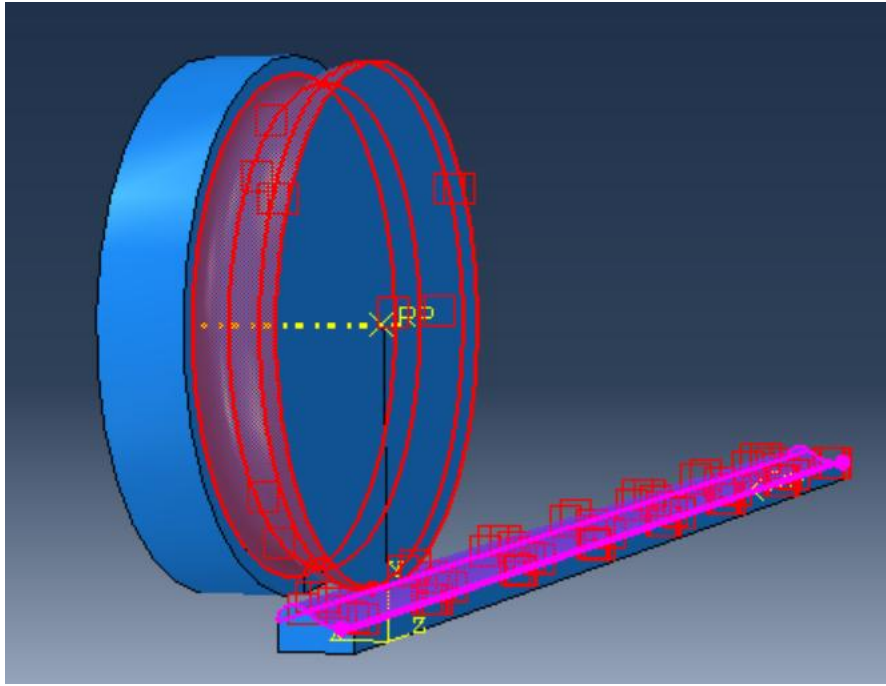


Figure 45: Surface contact interaction for constricted box model in Abaqus

The boundary conditions created are the same as the square-diamond model. A roller reference point is created at the centre of the roller which is used to define the movements of the roller. A compression boundary condition is created which specifies a 2 mm downward translation of the roller during the compression step to achieve the correct roll gap and does not allow any other translations or rotations for the roller. A rotation boundary condition is subsequently created which specifies a rotation of 6 radians around the axis marked in Fig.42 by the yellow dashed line. The compression boundary condition is updated during the rotation step and doesn't specify rotation about this axis to avoid conflict with the newly created rotation boundary condition. The degree of rotation is much larger than the previous models due to the much smaller roll radius used.

Two symmetry boundary conditions are created once again to ensure an accurate replication of the entire model. A y-symmetry boundary condition is created with the flat underside surface of the partial workpiece defined as the symmetry plane. An x-symmetry boundary condition is also created with the flat inner surface of the partial roller (visible in Fig.42) defined as the symmetry plane. The use of two symmetry planes once again ensures the workpiece can only move parallel to the z-axis and prevents any twisting or distortion of the model.

Creating Data

The data is recorded once again using history output requests. The roll force is recorded by creating a history output request for the reaction force RF2 (reaction force in the vertical direction) using the roller reference point as the domain. This produces a time dependant graph from which steady state rolling can be identified and the roll force obtained.

To obtain the maximum width of the outgoing workpiece, multiple sets are created consisting of individual nodes at the outermost edge of the undeformed workpiece. One of these nodes is labelled RP in Fig.41. History output requests are created for the displacement U1 (displacement in the direction perpendicular to rolling) for each of these nodes. This creates time dependant graphs for each node showing the lateral displacement throughout the entire rolling process. Multiple nodes are created to ensure that the widths obtained from the data are representative of steady state rolling geometries. Once the lateral displacement at one side is obtained, it is doubled to obtain the total increase in width and then added to the original width to obtain the maximum outgoing width.

Once the history output requests are finalised, a job is created and submitted to run the simulation.

4. Results

4.1. Flat Rolling Model

The flat rolling simulations are carried out only in PyRoLL and compared to the results of the Bekaert flat rolling reference tool. While not the main priority of this study, flat rolling simulations still follow many of the same fundamental principles as shape rolling and PyRoLL should be able to predict with relative accuracy the outcome of flat rolling processes if it is to be suitable for shape rolling simulations.

4.1.1. Roll Force

There are 6 roll forces predicted by PyRoLL consisting of three passes for each of the flat rolling simulations, the small and large diameter incoming wires. The predicted values for these roll forces are shown in table 7 alongside the outputs of the Bekaert reference tool. The difference between the PyRoLL prediction and Bekaert reference values are also shown as a percentage.

Table 7: Roll force results for flat rolling models

Roll Pass	Bekaert Reference Tool	PyRoLL Prediction	% Difference
3 mm flat wire – Pass 1	225.380 kN	540.216 kN	+ 139.69 %
3 mm flat wire – Pass 2	445.119 kN	582.498 kN	+ 30.86 %
3 mm flat wire – Pass 3	383.438 kN	511.221 kN	+ 33.33 %
14 mm flat wire – Pass 1	305.657 kN	705.503 kN	+ 130.82 %
14 mm flat wire – Pass 2	413.987 kN	348.862 kN	- 15.73 %
14 mm flat wire – Pass 3	432.782 kN	361.472 kN	- 16.48 %

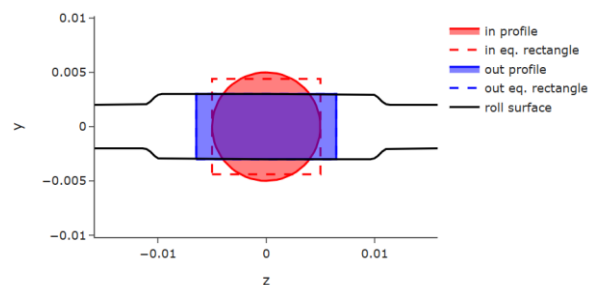
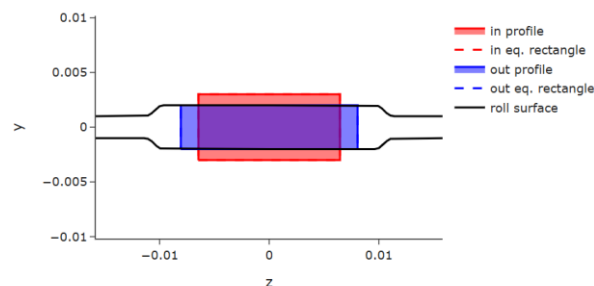
4.1.2. Geometry

The outgoing geometry is predicted by PyRoLL for all 6 passes like the roll forces. Table 8 shows the width of the outgoing cross section after each pass as predicted by the Bekaert reference tool, PyRoLL, and the percentage difference between the two predictions.

Table 8: Outgoing width results for flat rolling models

Roll Pass	Bekaert Reference Tool	PyRoll Prediction	% Difference
3 mm flat wire – Pass 1	13.002 mm	12.920 mm	- 0.63 %
3 mm flat wire – Pass 2	15.627 mm	16.123 mm	+ 3.17 %
3 mm flat wire – Pass 3	17.259 mm	17.969 mm	+ 4.11 %
14 mm flat wire – Pass 1	21.777 mm	21.471 mm	- 1.41 %
14 mm flat wire – Pass 2	22.297 mm	21.631 mm	- 2.99 %
14 mm flat wire – Pass 3	22.914 mm	21.834 mm	- 4.71 %

PyRoll also provides graphics of the predicted roll passes where the incoming cross section is represented by a red shaded area, the outgoing cross section is represented by a blue shaded area, the equivalent rectangle (for non-rectangular cross section) is represented by dashed lines of the same colour, and the roll surface is represented by black lines. The PyRoll cross section graphics for all six flat rolling passes are shown in Fig.46-51.

**Figure 46:** PyRoll cross section prediction for 3 mm flat wire model – pass 1**Figure 47:** PyRoll cross section prediction for 3 mm flat wire model – pass 2

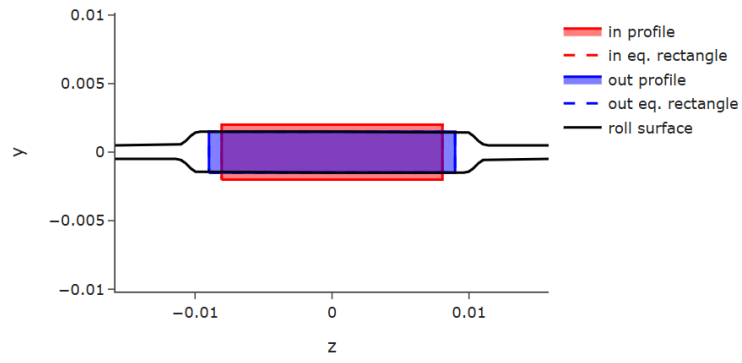


Figure 48: PyRoll cross section prediction for 3 mm flat wire model – pass 3

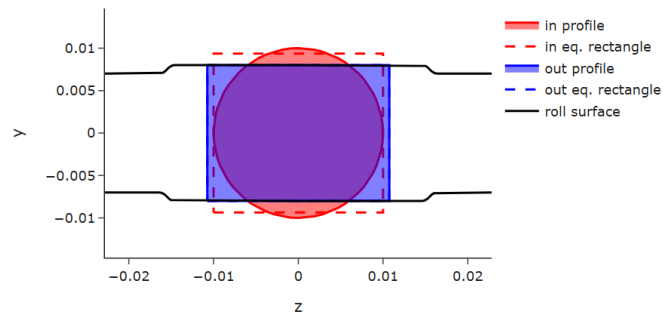


Figure 49: PyRoll cross section prediction for 14 mm flat wire model – pass 1

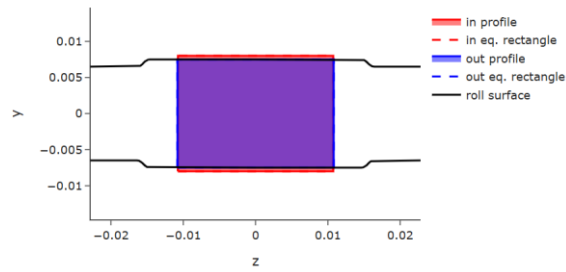


Figure 50: PyRoll cross section prediction for 14 mm flat wire model – pass 2

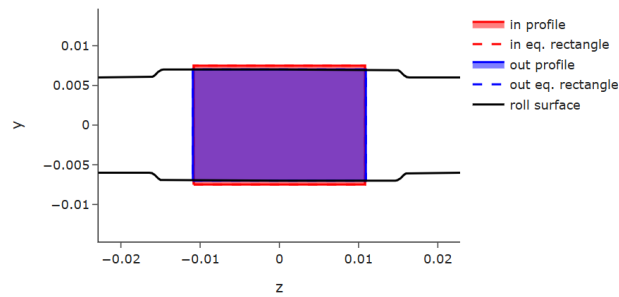


Figure 51: PyRoll cross section prediction for 14 mm flat wire model – pass 3

4.2. Round-Oval Model

4.2.1. Roll Force

Table 9 presents the roll forces values estimated by the experimental measurements and predicted by the PyRoIL and Abaqus tools along with the percentage error of these predictions against the experimental values. The time dependent graph produced during the FEA simulation is shown in Fig.52.

Table 9: Roll force results for round-oval models

Roll Model	Roll Force	Percentage Error
Experimental Reference	700-735 kN	N/A
PyRoIL	673.689 kN	- 3.76-8.34 %
Abaqus FEA	635 kN	- 9.29-13.61 %

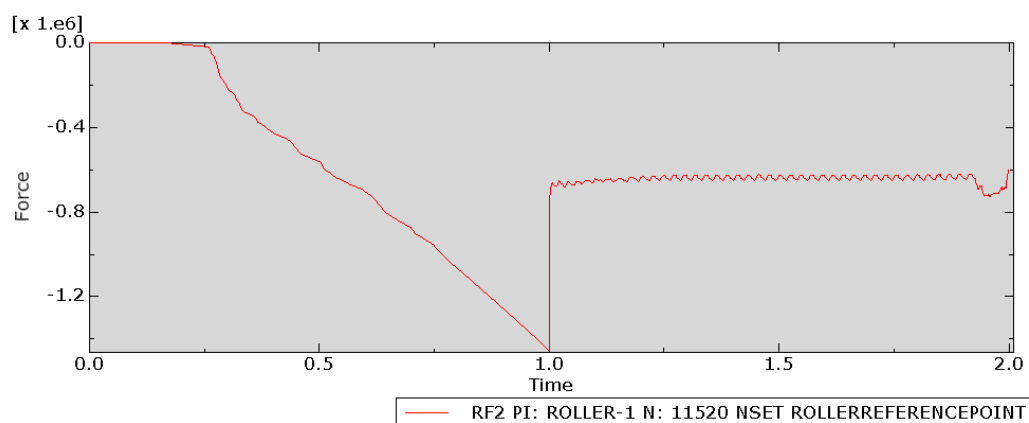


Figure 52: Abaqus roll force predictions with respect to time for round-oval model

4.2.2. Geometry

Table 10 presents the outgoing width values estimated by the experimental measurements and predicted by the PyRoIL and Abaqus tools along with the percentage error of these predictions against the experimental values. A time dependant graph produced during the FEA simulation of the lateral displacement of a node under steady state rolling is shown in Fig.53.

Table 10: Outgoing width results for round-oval models

Roll Model	Maximum Width	Percentage Error
Experimental Reference	68 mm	N/A
PyRoLL	67.15 mm	- 1.25 %
Abaqus FEA	65.54 mm	- 3.62 %

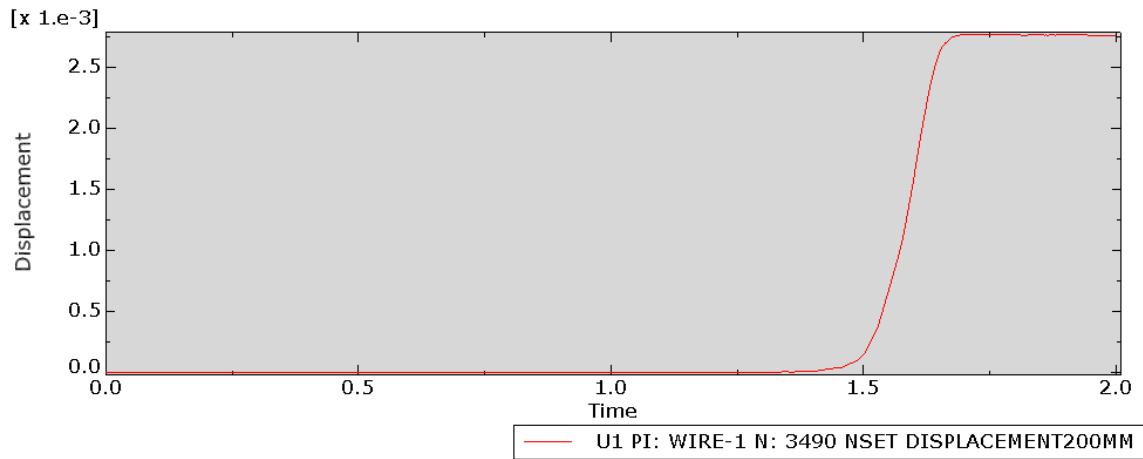


Figure 53: Abaqus node displacement predictions with respect to time for round-oval model

The cross sectional profiles predicted by PyRoLL and Abaqus FEA simulations are shown in Fig.54-58.

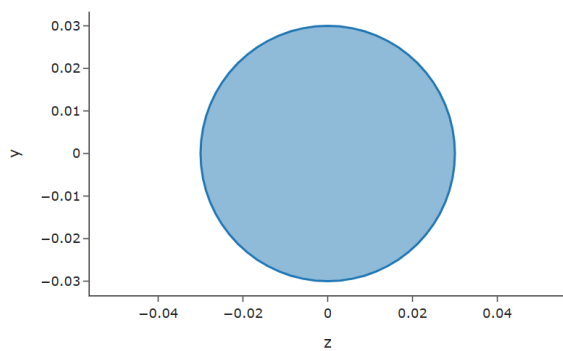


Figure 54: PyRoLL incoming cross section for round-oval model

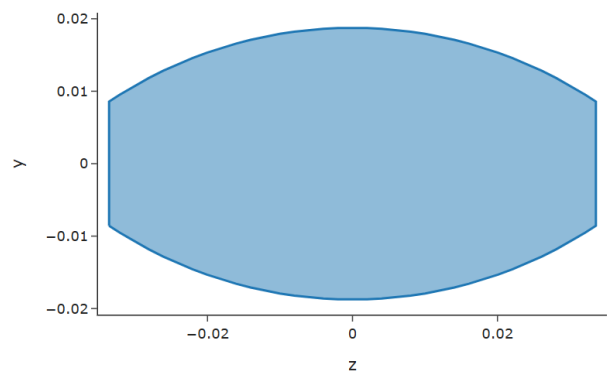


Figure 55: PyRoLL predicted outgoing cross section for round-oval model

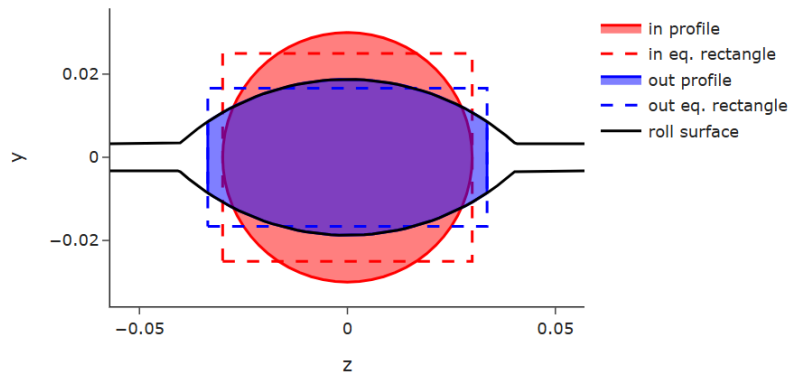


Figure 56: PyRoll cross section overlap for round-oval model

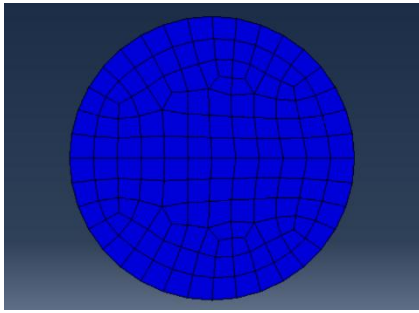


Figure 57: Abaqus incoming cross section for round-oval model

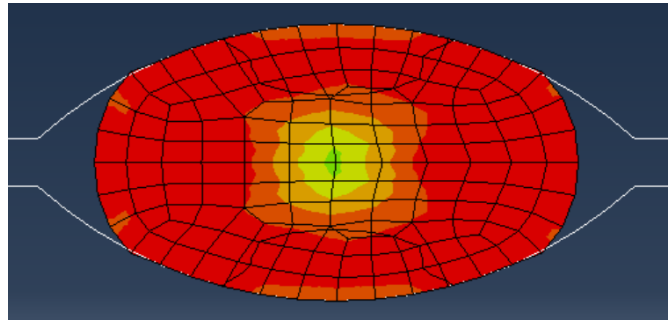


Figure 58: Abaqus predicted outgoing cross section for round-oval model

4.3. Square-Diamond Model

4.3.1. Roll Force

Table 11 presents the roll forces values estimated by the experimental measurements and predicted by the PyRoll and Abaqus tools along with the percentage error of these predictions against the experimental values. The time dependant graphs produced during the FEA simulations are shown in Fig.59-61.

Table 11: Roll force results for square-diamond models

Roll Model	Roll Force	Percentage Error
Experimental Reference	95 kN	N/A
PyRoll	108.071 kN	+ 13.76 %
Abaqus FEA – 2.0 mm Mesh	99.024 kN	+ 4.24 %
Abaqus FEA – 1.5 mm Mesh	95.733 kN	+ 0.77 %
Abaqus FEA – 1.0 mm Mesh	93.170 kN	- 1.93 %

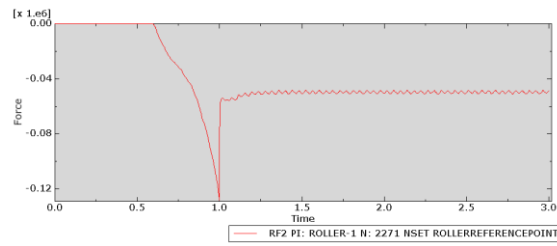


Figure 59: Abaqus roll force predictions with respect to time for square-diamond model (2 mm mesh)

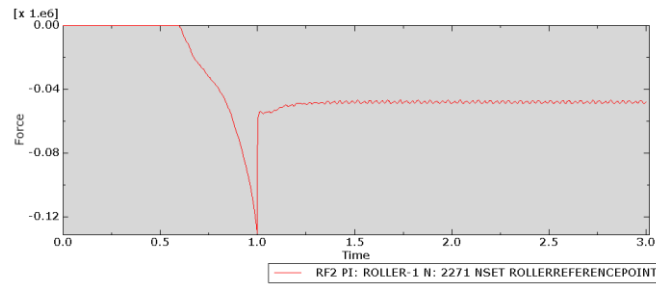


Figure 60: Abaqus roll force predictions with respect to time for square-diamond model (1.5 mm mesh)

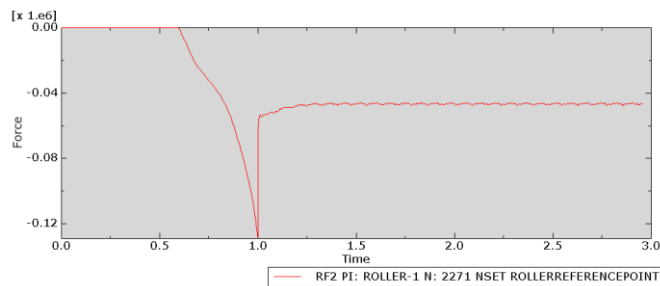


Figure 61: Abaqus roll force predictions with respect to time for square-diamond model (1 mm mesh)

4.3.2. Geometry

Table 12 presents the outgoing width values predicted by the PyRoLL and Abaqus tools. The time dependant graphs produced during the FEA simulations showing the lateral displacement of the outer nodes under steady state rolling are shown in Fig.62-64.

Table 12: Outgoing width results for square-diamond models

Roll Model	Maximum Width
PyRoLL	22.48 mm
Abaqus FEA – 2.0 mm Mesh	21.99 mm
Abaqus FEA – 1.5 mm Mesh	21.82 mm
Abaqus FEA – 1.0 mm Mesh	21.73 mm

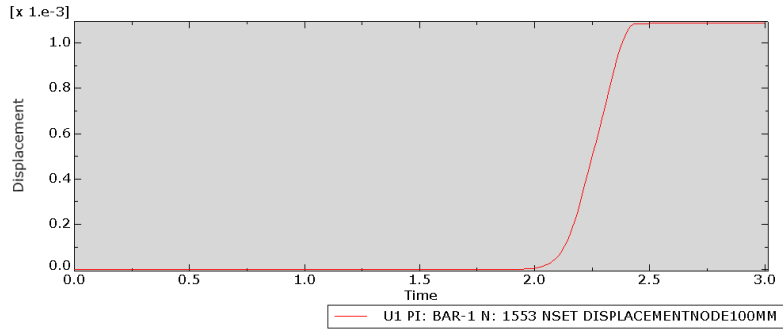


Figure 62: Abaqus node displacement predictions with respect to time for square-diamond model (2 mm mesh)

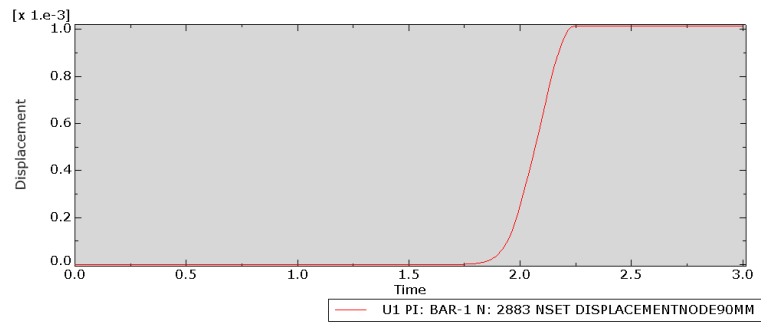


Figure 63: Abaqus node displacement predictions with respect to time for square-diamond model (1.5 mm mesh)

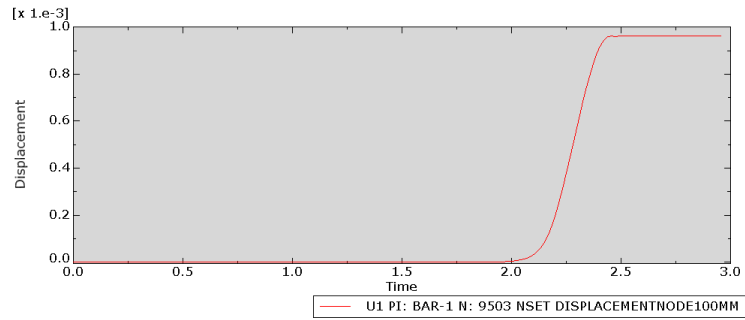


Figure 64: Abaqus node displacement predictions with respect to time for square-diamond model (1 mm mesh)

The cross sectional profiles predicted by PyRoLL and Abaqus FEA simulations are shown in Fig.65-73.

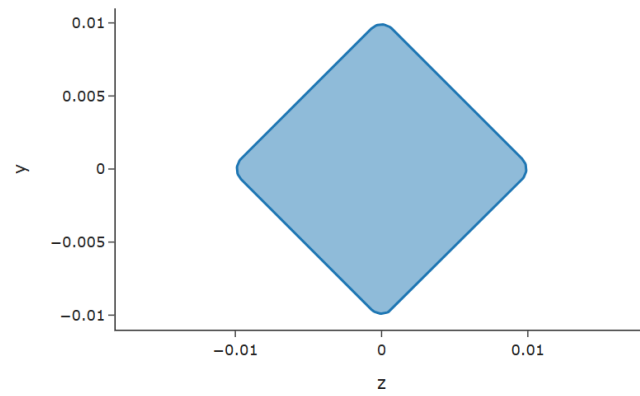


Figure 65: PyRoLL incoming cross section for square-diamond model

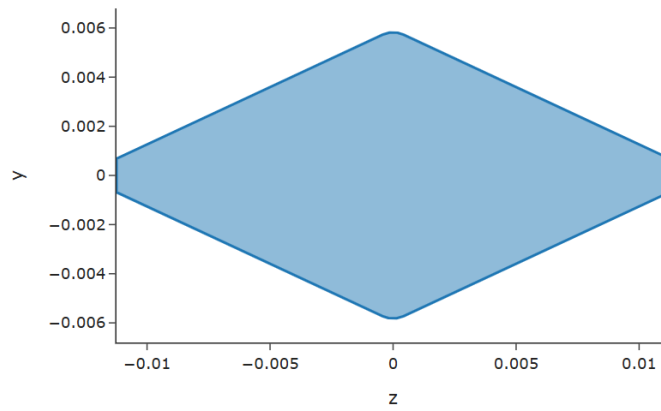


Figure 66: PyRoLL predicted outgoing cross section for square-diamond model

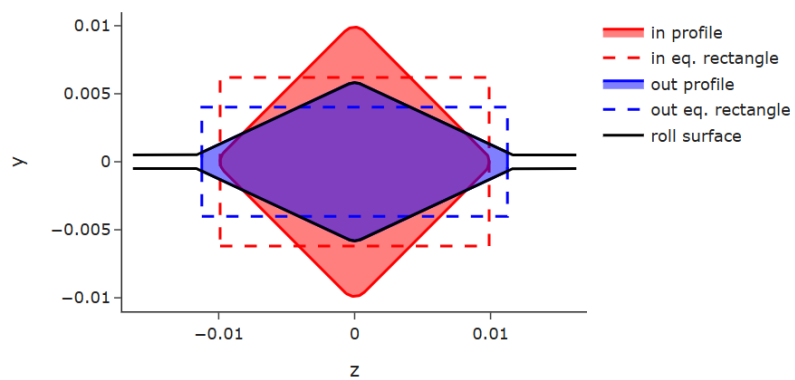


Figure 67: PyRoLL cross section overlap for square-diamond model

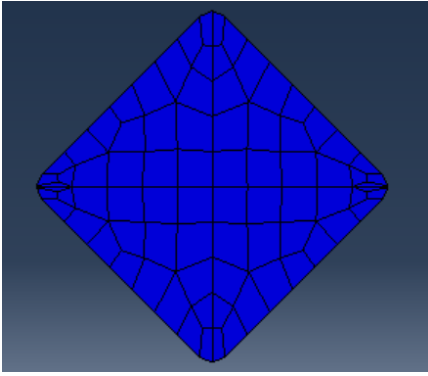


Figure 68: Abaqus incoming cross section for square-diamond model (2 mm mesh)

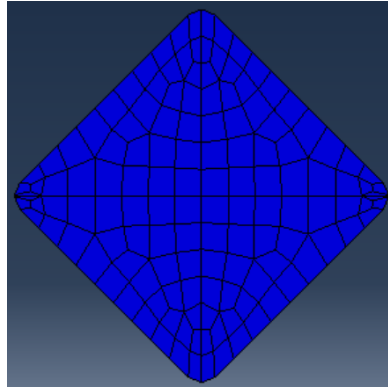


Figure 69: Abaqus incoming cross section for square-diamond model (1.5 mm mesh)

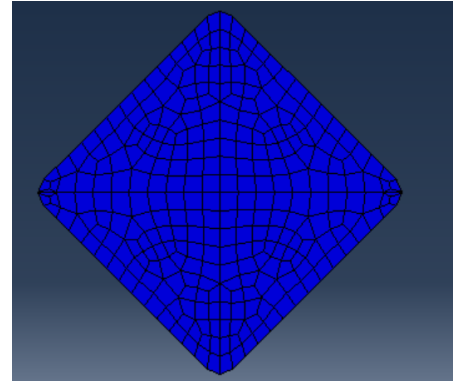


Figure 70: Abaqus incoming cross section for square-diamond model (1 mm mesh)

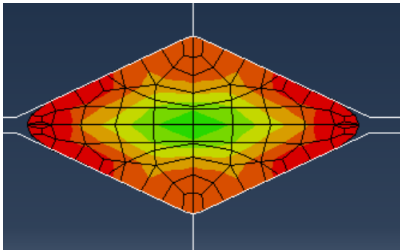


Figure 72: Abaqus predicted outgoing cross section for square-diamond model (2 mm mesh)

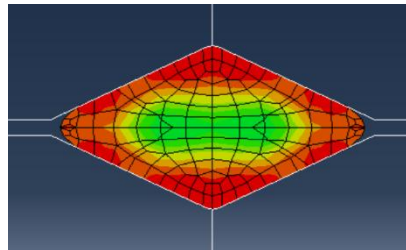


Figure 73: Abaqus predicted outgoing cross section for square-diamond model (1.5 mm mesh)

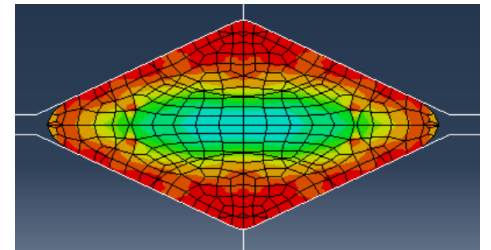


Figure 71: Abaqus predicted outgoing cross section for square-diamond model (1 mm mesh)

4.4. Constricted Box Model

4.4.1. Roll Force

Table 13 presents the roll force values predicted by the PyRoLL and Abaqus tools. The time dependant graph produced during the FEA simulation showing the roll force under is shown in Fig.74.

Table 13: Roll force results for constricted box models

Roll Model	Roll Force
PyRoLL	16.559 kN
Abaqus FEA	34.871 kN

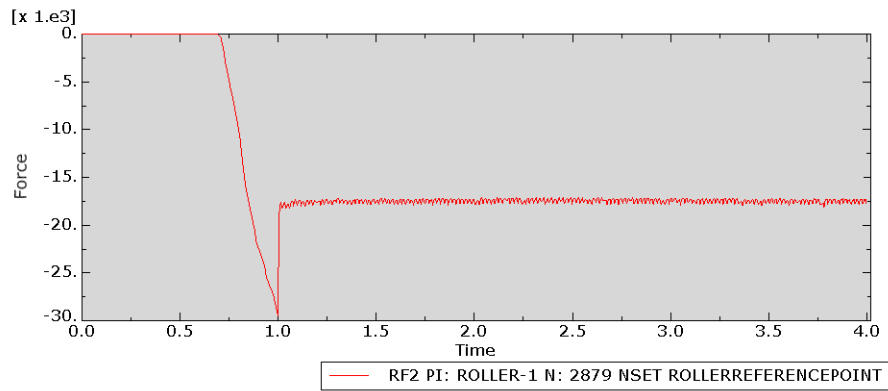


Figure 74: Abaqus roll force predictions with respect to time for constricted box model

4.4.2. Geometry

Table 14 presents the outgoing width values predicted by the PyRoll and Abaqus tools. The time dependant graph produced during the FEA simulation showing the lateral displacement of an outer node under steady state rolling is shown in Fig.75.

Table 14: Outgoing width results for constricted box models

Roll Model	Maximum Width
PyRoll	9.17 mm
Abaqus FEA	9.56 mm

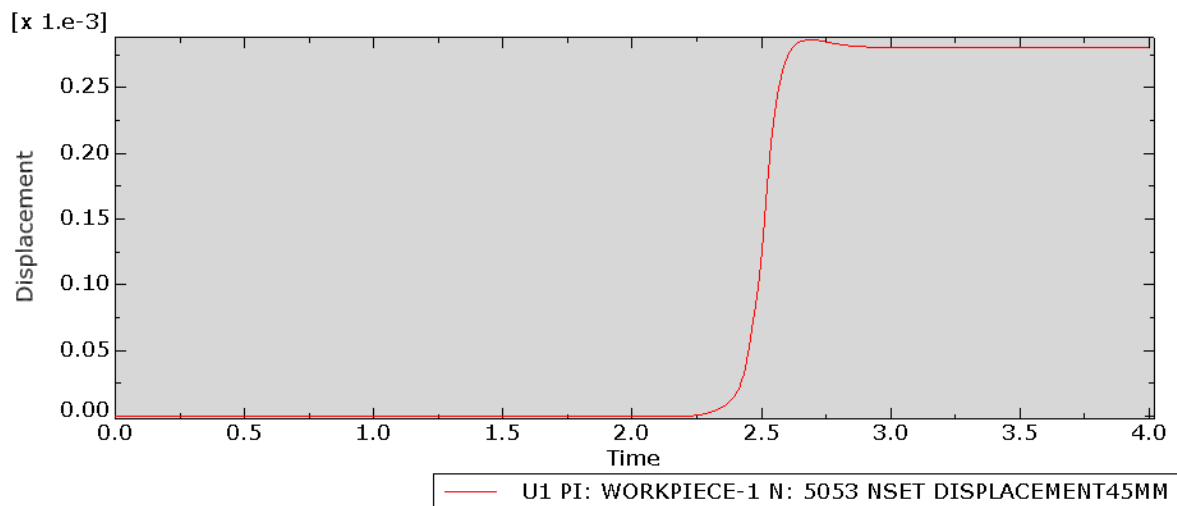


Figure 75: Abaqus node displacement predictions with respect to time for constricted box model

The cross sectional profiles predicted by PyRoll and Abaqus FEA simulations are shown in Fig.76-80.

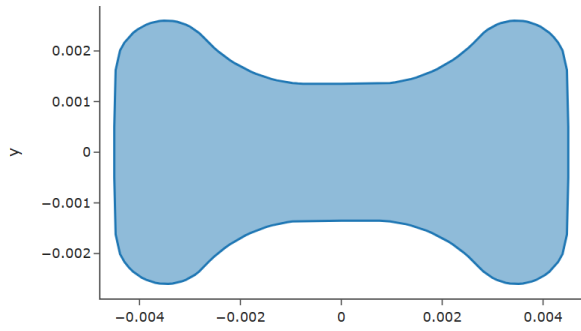


Figure 76: PyRoLL incoming cross section for constricted box model

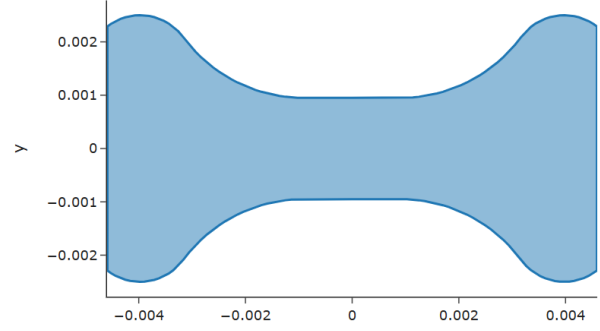


Figure 77: PyRoLL predicted outgoing cross section for constricted box model

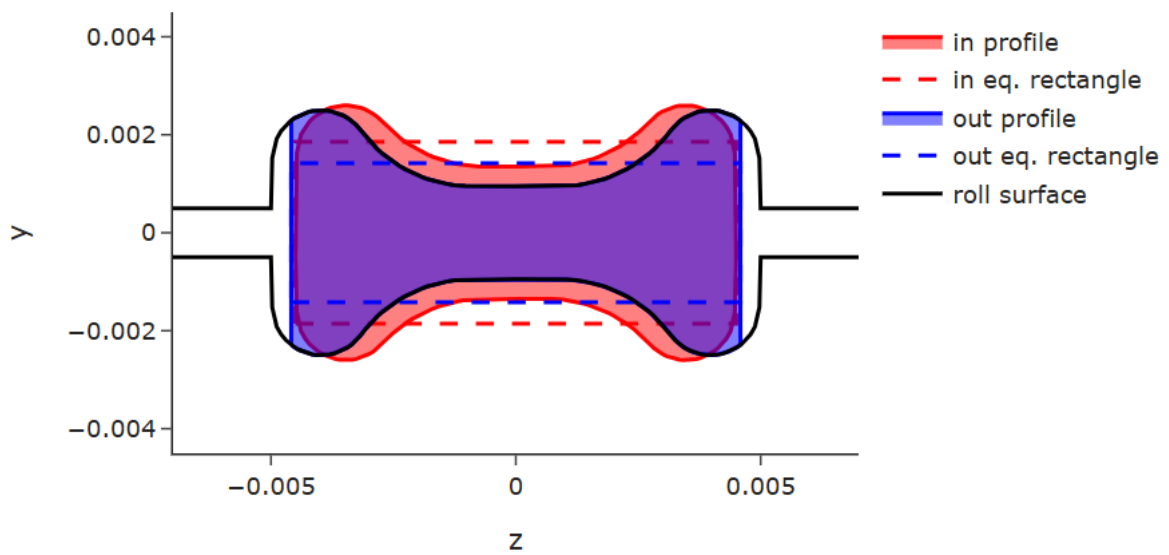


Figure 78: PyRoLL cross section overlap for constricted box model

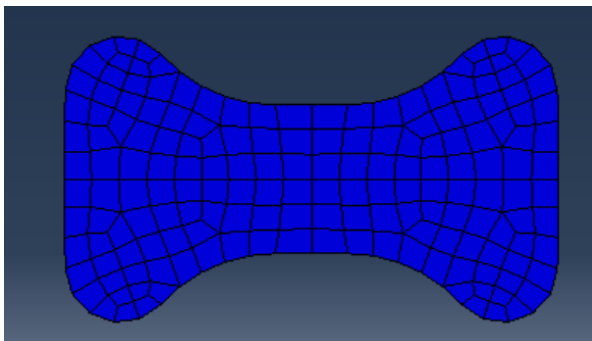


Figure 79: Abaqus incoming cross section for constricted box model

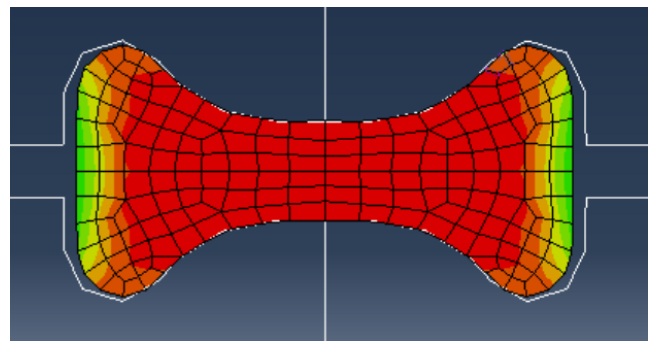


Figure 80: Abaqus predicted outgoing cross section for constricted box model

5. Discussions

5.1. Flat Rolling Model

The flat rolling model provides a good starting point for assessment of the PyRoLL model. The first comparison with the Bekaert model is the roll force which shows quite a poor correlation. For the first pass of each sequence, PyRoLL overpredicts the roll force by over 130 % but for the second and third passes it predicts roll forces much closer to the Bekaert reference values. For the smaller diameter wire it overpredicts the second and third passes by just over 30 %, while for the larger wire it underpredicts by just over 15 % when compared to the reference tool.

It must be noted that the results available from this reference tool are effectively coming from a black box and the method is unknown. All that is known is the results provide a suitable representation of the real world outcome for it to be used in industry but the PyRoLL model can only be developed based on the limited data available. The PyRoLL model does not account for conditions which may have effects on the forces experienced by the rolls. The Bekaert tool does not for example, specify any front or back tension that is used in this simulation but spools before and after the rolls do provide tension in real scenarios which can cause deformation to a degree. This tension is not considered in the PyRoLL model simply because the information isn't available to provide a suitable input. The material hardening model is also made by fitting a complex general formula to the 4 available data points which may work in some cases but is not a reliable method of developing models. In order to make a representative material hardening model in PyRoLL, the actual material hardening properties should be known and quantified or at the very least, a significantly larger number of data points should be available to fit the data strain hardening relationship appropriately.

While the lack of reference data does make it difficult to draw conclusions from the flat rolling model with regard to roll force, a point of note is that the discrepancy in roll force between PyRoLL and the reference tool was significantly larger for the first pass, which as seen in Fig.46 & Fig.49, consists of a round cross section passing into a flat roll. This indicates that PyRoLL may struggle more with certain geometries or combinations of geometries and that this should be considered in the testing of shaped geometries.

While the roll force provided a poor correlation, the outgoing geometries actually provided promising results. Despite the lack of reference data, the outgoing width after all six passes was within 5 % of the reference values with the predictions of the first passes differing by only

0.63 % and 1.41 %. Similarly to the roll force predictions, it must be noted that the lack of data makes it difficult to draw strong conclusions but it does provide a promising start that PyRoll can predict geometries within this level of accuracy to a tool used in industry with only limited reference inputs.

5.2. Round-Oval Model

The discrepancies between the PyRoll prediction, FEA predictions and experimental measurements are noticeable but not excessive, with both simulations underpredicting the roll force and outgoing widths. When compared to the experimental results, the PyRoll model underpredicted the width by 1.3 % and underpredicted the roll force somewhere in the range of 3.8-8.3 %. The FEA simulation showed larger discrepancies with the experimental results with width being underpredicted by 3.6 % and force being underpredicted somewhere in the range 9.3-13.6 %.

These discrepancies may be in part due to information about the experimental setup that had to be estimated. The results were not explicitly given but rather, the results of the model proposed by the authors were given as ratios of the experimental measurements. This means the results had to be calculated based on the proposed model using data that was only partially available with some data needing to be obtained elsewhere or estimated. Once the force was calculated from the proposed model, this value had to be converted to the actual measured forces based on the error ratios given in the paper which ranged 0.8-5.7% on average. Using this method, it is reasonable to estimate the experimental value is in the range of 700-735 kN but this is a large range and assuming that the authors used exactly the same inputs in their calculations.

It is also possible that there were differences between the created models and the actual physical experiment that took place. One example of this is the yield stress value used in the PyRoll and Abaqus simulations. A yield stress of 200 MPa is used which was obtained using a model specifically referenced in the paper but this model is given in the form of a graph with no method of calculating the yield stress for specific conditions. Another paper researching the effect of temperature on the strength of AISI 1018 steel (Badaruddin et al., 2020) suggests that the yield stress should be less than 100 MPa at 800°C. The selected yield stress of 200 MPa is the most reasonable estimate but ultimately it is still an estimate as the exact value used by the authors is not specified and there isn't enough information about the steel used to say for definite that the yield stress is equal to 200 MPa at this temperature.

Although there may be unknown discrepancies between the experimental and simulated results, the exact same inputs are used for both the PyRoIL and Abaqus simulations so discrepancies between the predictions should be explainable and also repeatable for similar but not identical models.

PyRoIL predicts a slightly larger roll force than Abaqus for this model which can be explained by a larger contact area. At its most basic level, the roll force comes down to a relationship with the materials resistance to deformation and the contact area upon which the roller exerts force. Both the PyRoIL and Abaqus models use the same yield stress but the contact area is different. Looking back at Fig.55 and Fig.58 which show the outgoing cross sections of the PyRoIL and Abaqus simulations respectively, a very noticeable observation is that PyRoIL predicts a flat outer edge. This would result in the contact patch between the roll and workpiece extending all the way to the outer edges of the width. The Abaqus cross section shows a rounded edge which suggests a narrower cross section is predicted that won't extend to the widest point of the cross section.

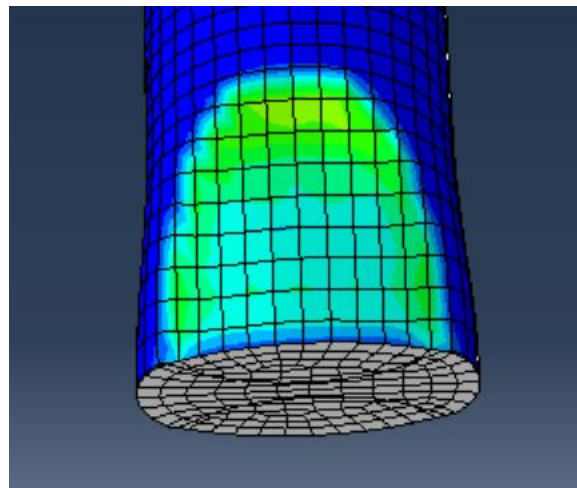


Figure 81: Abaqus predicted contact pressure zone for round-oval model

This is reinforced by Fig.81 which shows the contact area predicted by Abaqus. The combination of this contact area not extending to the outermost point and also the generally narrower cross section predicted by Abaqus, results in a smaller contact area and a lower roll force as a result.

PyRoIL also predicts a larger width than Abaqus. This increase in width can most likely be attributed to the equivalent rectangle method used to calculate deformation. The approximation does prove to be quite reasonable in this case with an underprediction of only 1.25% compared

to the experimental results and quite close to the prediction made by Abaqus as well but ultimately, it is still an approximation.

Finally, the actual geometry shape differs between both the Abaqus and PyRoll simulations as seen in Fig.55 and Fig.58. The simulation run in Abaqus predicts a rounded edge of the profile while PyRoll assumes a flat edge. The PyRoll prediction is based on the assumption that the side is completely flat with a 100% groove filling up to the maximum width. While this can provide reasonable predictions for the maximum outgoing width, it doesn't accurately represent the outgoing shape. The rounded edge predicted by Abaqus can be explained by the presence of sticking friction between the rollers and the workpiece which occurs when the shear stress imparted on the workpiece is greater than the shear strength of the material. While the workpiece sticks to the groove surface, shear deformation within the metal occurs and the bulk material spreads more than the contact surface and adjacent material, resulting in a wider cross section along the centre line of the workpiece than at the contact surface.

5.3. Square-Diamond Model

For this model, a mesh sensitivity analysis was carried out on the Abaqus model with workpiece mesh sizes of 1 mm, 1.5 mm, and 2 mm. The general trend observed is a convergence towards lower values for both roll force and outgoing width with decreasing mesh size. The outgoing width only varied by 0.2 mm across the entire tested mesh sizes. The roll forces did show a slightly larger variation but still not a significant difference, ranging from 93 kN to 99 kN. It is also apparent by looking at the time dependent graphs for the roll forces (Fig.59-61) that the smaller mesh sizes produce less noise around steady state rolling. This does instil more confidence when specifying a value for the roll force obtained from the steady state portions of the force graphs.

The same trends observed in the round-oval model are seen in this model when comparing the PyRoll to Abaqus results, regardless of the mesh size. PyRoll predicts a slightly larger roll force and maximum width compared to Abaqus, even with the largest tested mesh size. The PyRoll prediction for width is only 0.5-0.7 mm larger than the Abaqus predictions and there is a slightly larger discrepancy with force ranging from 9-15 kN difference, once again PyRoll predicting the larger value. The larger predictions from PyRoll can be explained by the same mechanisms as the round-oval model. The equivalent rectangle estimation will cause some discrepancy in maximum width, leading to the larger PyRoll prediction. Sticking friction results in Abaqus predicting a rounded edge in Abaqus (Fig.71-73) which is assumed to be flat

in PyRoIL (Fig.66). The rounded edges lead to a contact area that doesn't extend to the outer edges (Fig.82-84) which combined with the generally narrower cross section, results in a smaller prediction for roll force. The contact area doesn't vary much with mesh size but the edges are sharper with smaller mesh sizes.

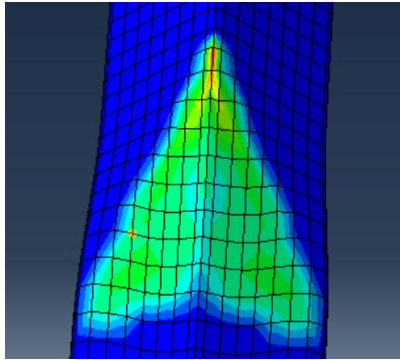


Figure 82: Abaqus predicted contact pressure zone for square-diamond model (2 mm mesh)

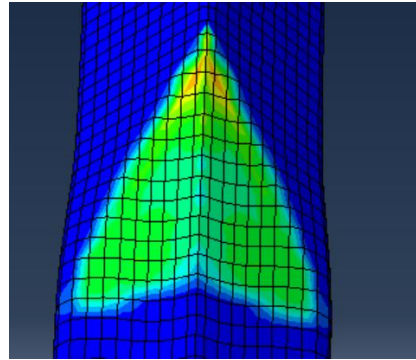


Figure 83: Abaqus predicted contact pressure zone for square-diamond model (1.5 mm mesh)

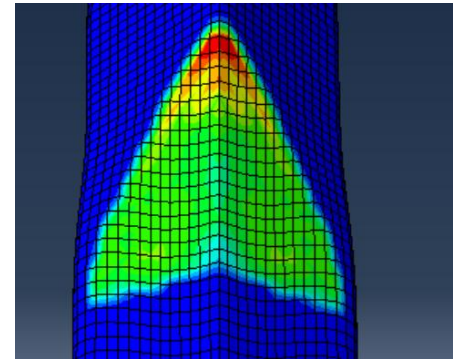


Figure 84: Abaqus predicted contact pressure zone for square-diamond model (1 mm mesh)

While the same trends are seen between the PyRoIL and Abaqus simulations, the trends are reversed when compared to the experimental results. Unfortunately, data isn't available for the experimental width but the roll forces predicted by all mesh sizes in Abaqus were closer to the experimental result than PyRoIL, unlike the round-oval model where PyRoIL made closer predictions. Once again, this doesn't necessarily mean that Abaqus simulations are more accurate as there may be discrepancies between the experimental setup and the simulations which had to be estimated based on the available data.

5.4. Constricted Box Model

This model shows very different results compared to the previous models and means the theories surrounding the results need to be revisited and adjusted or changed. PyRoIL predicted a slightly larger width and roll force for the previous models than Abaqus but in this model, it predicted a slightly narrower cross section and a force of less than half that predicted by Abaqus. Unfortunately, there is no experimental data to compare this to but the discrepancies between the two simulations can be explained.

The cross sectional shape differs once again with PyRoIL showing a flat edge (Fig.77) while Abaqus predicts a round edge (Fig.80). This can be attributed to the sticking friction that is not accounted for in PyRoIL. The narrower width predicted by PyRoIL may be due to the unsuitability of the equivalent rectangle method to estimate deformation for a geometry like this. That method was designed for use in more standard geometries such as the round-oval and

similar geometries like the square-diamond. It is simply not designed to work with geometries such as this which include an indent or at least for large indents such as the one in this model. It is also possible that due to the nature of the geometry of the groove shape, the component of the normal force in the direction perpendicular to rolling actually pushes the material outwards and serves to increase the spread of the workpiece which wouldn't be the case with the previous geometries.

The roll force predicted by PyRoLL being less than half that predicted by Abaqus is a very unusual result. A smaller force could be explained by the narrower cross section predicted by PyRoLL giving a smaller contact area and actually reinforcing the observation for the previous models but the difference in width is very minor and wouldn't cause this much of a gap in roll force. Looking at the contact pressure profile in Abaqus (Fig.85) reveals a difference to the previous models. The contact patches in the previous simulations showed relatively uniform contact pressures across the contact area with small zones of high pressure but nothing significant. Looking at this contact patch shows a different story with a contact pressure largely concentrated in the centre of the profile. The green area around the edge of the contact is representative of a pressure approximately the same as the yield stress but the red section in the centre is actually over double the yield stress. PyRoLL on the other hand assumes a uniform contact pressure based on the yield stress which gave similar results to Abaqus previously when it also predicted a broadly uniform contact pressure. This non-uniform contact pressure for the constricted box could explain the significant difference in roll force without discrediting the observations of the previous models.

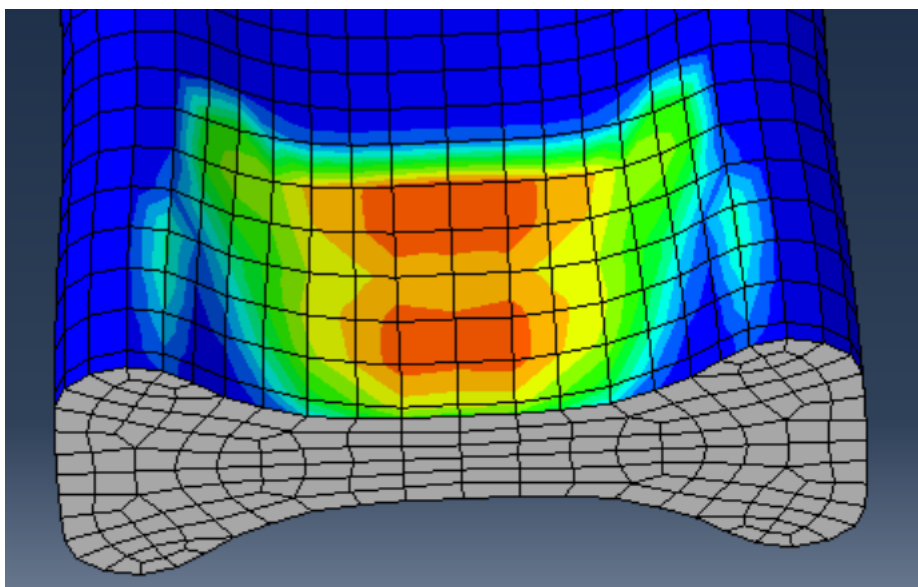


Figure 85: Abaqus predicted contact pressure zone for constricted box model

While the round-oval and square-diamond models provided relatively close predictions between Abaqus and PyRoLL which also both came within acceptable levels of experimental measurements based on the available data, the same cannot be said for this geometry. It can't be definitively said that Abaqus is correct for this model as there are no experimental measurements to back it up but based on the simplifications made in PyRoLL and the data seen in Abaqus, it is most likely that Abaqus is at least a more suitable representation of this model.

6. Conclusions and Future Work

6.1 Conclusions

A number of conclusions can be made from the results obtained in this study. PyRoLL certainly proved to be a much more efficient method of simulating advanced rolling processes. The models in PyRoLL took less than 10 seconds each to run on a basic laptop whereas the FEA simulations run in Abaqus took hours to run on a powerful PC. The difference in time taken to run the simulations was so large that they can't be compared. On top of the actual simulation time, the model development was significantly quicker in PyRoLL as well. A full model can be designed from start to finish in PyRoLL in a matter of minutes due to the modular design. The parameters are selected and defined very easily in PyRoLL and the tool then creates the simulation without any further input whereas Abaqus requires a lot of model development from scratch. The creation of the parts is quicker in PyRoLL as a small number of parameters are defined rather than sketching the profile from scratch but the actual rolling setup is where PyRoLL is much quicker. Once the parts and rolling parameters are defined, PyRoLL creates the simulation automatically whereas Abaqus requires almost every aspect to be specified. The parts had to be located in exactly the right position with respect to each other, the time steps to be defined, boundary conditions specified to perform the rolling process along with extra conditions to ensure the simulation runs smoothly, contact interactions defined between parts, and even requests for the output data to be recorded. All in all, the PyRoLL model creation is significantly quicker but also due to the simplicity, there is much less potential for errors when creating the model.

PyRoLL appeared to predict larger roll forces and outgoing widths when compared to Abaqus, at least for the standard geometries of the round-oval and square-diamond models. The widths can be explained by the equivalent rectangle estimations which produced reasonable results but are still estimations so ultimately would be expected to differ from the FEA predictions. The forces can be explained by a larger contact area caused by the generally wider cross section and the flat edges assumed by PyRoLL, resulting in an even wider contact area. These explanations should ensure repeatable results and that PyRoLL would predict larger forces and widths than Abaqus for any model of similar geometry.

While the numerical width and forces are comparable between PyRoLL and Abaqus, the actual shape of the outgoing profile isn't. PyRoLL assumes a flat side for all roll passes whereas Abaqus predicts rounded edges. The rounded edges can be explained by the presence of

sticking friction between the workpiece and rolls causing the centre of the workpiece to spread more than the surface in contact and this is also seen in the experimental results. This means regardless of the accuracy of the maximum width values, the profile predicted by PyRoll will not be 100% accurate for any simulation.

The constricted box geometry did not repeat the trends of PyRoll predicting slightly larger widths and forces but instead PyRoll predicted a slightly smaller width and significantly smaller force than Abaqus. Based on these results, it does not look like PyRoll is suitable for more irregular geometries. The models and assumptions are just too simplified to reflect the real world behaviours of geometries like this. The equivalent rectangle method is a major simplification of the calculations and it is designed for more regular geometries like the round-oval model rather than this type of shape with an indent. The PyRoll assumption of uniform contact pressure is also a significant simplification. In the round-oval and square-diamond this assumption turned out to be quite similar to the prediction from Abaqus but not for the constricted box model. I believe if PyRoll is to be used for more complex geometries, the models will need to be changed to remove some of the simplifications/assumptions or adjust the models to specifically suit the geometries available to test.

PyRoll did compare well to Abaqus but ultimately, it is meant to recreate the real world process, not the Abaqus output. Based on the experimental data available, PyRoll performs well. The force and width values are within an acceptable range of the experimental measurements. The widths predicted for the flat rolling validation were also very close to the expected reference values despite being based on very limited information to recreate the model. The roll forces were very different in the flat rolling but as mentioned, the data available was very limited. For the shape rolling models, PyRoll predicted forces and widths closer to the measured values than Abaqus on the round-oval simulation while Abaqus predicted closer on the square-diamond simulations. Based on this data, PyRoll can't be declared as definitively better or worse than the Abaqus simulations but based on the available experimental data, PyRoll produced results that are just as good as the FEA models with much greater efficiency.

These observations leave the question of what can PyRoll be used for? It appears that PyRoll is unsuitable for complex geometries, at least in its current state but it shows a lot of potential as an efficient alternative to FEA for more regular geometries like the round-oval and square-diamond shapes. The larger predictions for the roll forces could be beneficial in roll pass design as they might provide a larger margin of safety. A simulation could be run much quicker in

PyRoLL than in Abaqus to determine the roll force and if it is known that the prediction will be slightly higher than Abaqus, the machine will be less likely to fail due to overloading.

PyRoLL does not have the same potential for the outgoing geometry as a sole model prediction, at least in its current state. Slightly larger predictions for the width aren't necessarily better as it could push the final product to be further away from the design specification. This in combination with the flat edges that don't accurately represent the outgoing shape mean PyRoLL can't be used on its own to predict the outgoing profile for a rolling process. It does however have potential for use as an initial trial and error tool in roll pass design. The predicted geometry may not be exactly correct but it has repeatedly predicted widths quite similar to both experimental and FEA results. PyRoLL can't predict exact geometries but it can get very close in a short timeframe. From here, the model can be validated with a single FEA simulation or physical prototype as opposed to developing multiple FEA simulations or prototypes to find a suitable solution.

6.2 UN SDGs

The use of analytical and empirical models such as PyRoLL have the potential to improve the rolling industry but they also have the potential to contribute towards the United Nations Sustainable Development Goals. The implementation of tools such as this would help to achieve higher levels of economic productivity through innovation and reduced time spent on simulations and contribute to goal number 8 which aims to promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

6.3 Future Work

The PyRoLL tool has potential to positively influence the industry but it isn't perfect. If I were to continue this project, I would like to carry out some experimental testing of my own. There is limited data available in literature which provides both rolling parameters to replicate in a simulation and also output data to compare the results with. Physical testing to go alongside the simulation testing of PyRoLL would provide more accurate input data to create the models, more reliable results to compare with the simulations and increased flexibility to test different models to see how specific parameters can affect the results or to test a wide variety of different geometry shapes.

On top of additional validation testing for the regular geometry shapes, new empirical models could be developed that are more suitable for the less regular geometry shapes such as the

constricted box. This would provide a wider variety of applications that PyRoLL can be used for with confidence that the results are representative of real world rolling processes.

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Appendix A: Calculations for Round-Oval Experimental Results

The formula for roll force is given by Eq.16 where P represents the average contact stress applied to the workpiece and A_d is the projected contact area. The predicted projected contact area is given in the paper as 2843 mm². This is actually an underprediction when compared to the measured contact area but since the predicted force is being calculated instead of measured, all values must be taken as predicted by the model.

$$F = PA_d \quad (16)$$

An equation for the average contact stress in plate rolling is given by Eq.17 based under plane strain conditions. In this formula, K_m is the average deformation resistance of the workpiece, μ is the coefficient of friction, L is the projected contact length between the roll and workpiece, and h_m is the mean height of the workpiece. The authors have proposed a modified version of this equation, given in Eq.18, for rod geometries where the cross sections are adjusted to equivalent rectangles of equal area and equivalent contact lengths and mean heights are used based on these equivalent cross sections. It also includes a linear compensating function given by $(1-\varepsilon_1)$ which accounts for the fact that shape rolling can be analysed under a ‘weak plane-strain deformation condition’, where ε_1 represents the strain of the equivalent cross sections in the lateral spreading direction.

$$P_{plate} = K_m e^{\frac{\mu L}{2h_m}} \quad (17)$$

$$P_{rod} = (1 - \varepsilon_1) K_m e^{\frac{\mu \bar{L}}{h_m}} \quad (18)$$

The deformation resistance is characterised based on Shida’s Equation (Richard Shulkosky et al., 2005). The equation itself is not given but a graph plotting this model is available and shown in Fig.86. It is reasonable to estimate that the deformation resistance used in the model prediction was 200 MPa given that the metal contains 0.1% carbon and the test is carried at 800°C. This is used for the force calculations but is also required for the PyRoIL and Abaqus simulations. For these simulations, it is assumed that no strain hardening takes place considering that the rolling takes place well above the recrystallisation temperature for steel. The coefficient of friction is specified in the paper as being 0.35 and will be taken as such. The remainder of the values are based on geometry and the cross sections must be converted to equivalent rectangles first.

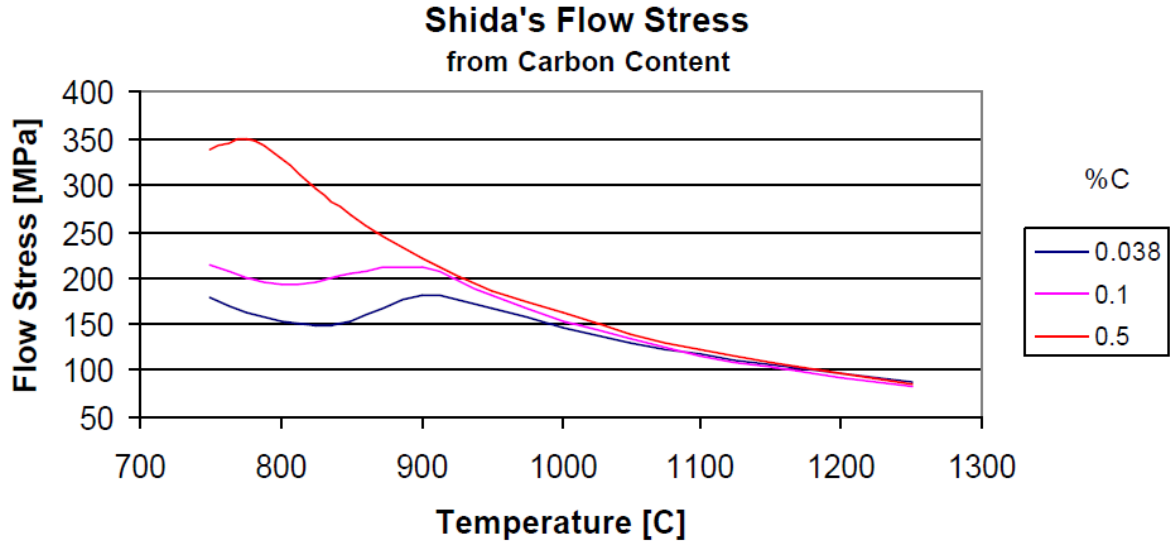


Figure 86: Shida's Flow Stress model used in estimating yield stress for round-oval and square-diamond models

The literature doesn't specify the exact method used to calculate the equivalent rectangles of the incoming and outgoing cross sections but it does specify that the area must be equal to the actual area and also gives a formula to calculate the strain ε_1 given by Eq.19 where $\bar{W}_i$ and $\bar{W}_p$ represent the widths of the incoming equivalent rectangle and the pass equivalent rectangle respectively.

$$\varepsilon_1 = \ln \frac{\bar{W}_i}{\bar{W}_p} \quad (19)$$

Looking back at Fig.8, the value for ε_1 is said to be 0.137. The incoming cross sectional area can be easily calculated as it is circular in shape and the predicted oval cross sectional area is 2103 mm² as given in the paper. Using this information and the assumption that the equivalent rectangle for the circular cross section is a square, the equivalent heights and widths can be calculated for both the incoming and outgoing cross sections and are given in Table 15.

Table 15: Geometrical parameters for cross sections of experimental round-oval model

	Incoming Cross Section	Roll Pass Cross Section
Cross Sectional Area	2830 mm ²	2103 mm ²
Actual Height	60 mm	37.5 mm
Actual Width	60 mm	68 mm
Equivalent Rectangle Height	53.20 mm	34.47 mm
Equivalent Rectangle Width	53.20 mm	61.01 mm

The average equivalent height is the average of the two equivalent heights and comes out to 43.85 mm. The effective projected length is calculated from Eq.20 which is given in the literature where G is the roll gap (6.5 mm) and R_{max} is the maximum roll radius (155 mm). The remaining variables are taken from the equivalent rectangle calculations. This effective projected length comes out to be 51.39 mm.

$$\bar{L} = \sqrt{\left\{R_{max} - \frac{1}{2}(\bar{H}_p - G)\right\}(\bar{H}_i - \bar{H}_p)} \quad (20)$$

All the relevant values are subbed into Eq.3 after dimensions of length are converted to metres and the average contact stress comes out to be 260 MPa and when put back into Eq.1 gives a vertical force of 740 kN.

Based on these calculations, it is reasonable that the proposed model predicts a force of 740 kN under the given conditions. The model is said to overpredict with a difference in the range of 0.8% to 5.7%. Using these errors, the experimental forces can be estimated from approximately 700 kN (5.7% error) to 735 kN (0.8% error).